

A Parametric Geometric Framework for Genetic Optimization of Single Bell Nozzle Contours

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Abstract

This work presents a parametric geometric framework for the generation, analysis, and preliminary optimization of single bell nozzle contours for hybrid rocket propulsion systems. The proposed formulation defines the nozzle wall as a continuous piecewise contour composed of a converging straight section, a subsonic circular throat arc, a supersonic circular throat arc, and a parabolic divergent section. The resulting geometry is fully determined by a reduced set of physically meaningful design parameters, including throat radius, expansion ratio, chamber radius, converging angle, initial supersonic angle, reference conical half-angle, and bell length fraction.

The framework combines classical quasi-one-dimensional isentropic nozzle relations with thermochemical input data to estimate axial Mach number, pressure, temperature, density, and velocity distributions along the generated contour. In addition to two-dimensional profile generation, the method supports three-dimensional reconstruction through surface revolution and coordinate export for CAD-based manufacturing workflows.

The formulation is coupled to a real-valued genetic algorithm that maximizes an effective ambient thrust coefficient. The implemented objective combines the RocketCEA momentum thrust coefficient with an exit-divergence correction, compressible boundary-layer blockage, integrated wall-shear loss, and the signed ambient pressure-thrust contribution. Expansion ratio, bell-length fraction, and initial divergent-wall angle are optimized, whereas throat size, chamber diameter, convergent angle, and reference cone half-angle remain fixed pre-sizing inputs. A BLIMP-inspired Cebeci–Smith profile marcher is used for final evaluation, while a weaker integral closure is retained for rapid screening. The framework is not intended to replace high-fidelity CFD or experimental qualification, but to provide a fast, interpretable preliminary design tool whose assumptions and loss contributions remain explicit.

Keywords— bell nozzle; hybrid rocket engine; parametric geometry; nozzle contour optimization; genetic algorithm; isentropic flow; boundary layer; CAD generation; propulsion design.

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1 Introduction

Rocket nozzle geometry is one of the dominant factors controlling the conversion of chamber thermal energy into directed exhaust kinetic energy. For a fixed chamber pressure, propellant mass flow rate and propellant combination, the nozzle determines the exhaust Mach number, exit pressure, exit velocity, thrust coefficient and degree of axial momentum alignment. An efficient nozzle must therefore expand the combustion gases as much as possible while avoiding excessive length, excessive divergence losses, flow separation, shock formation or manufacturing complexity.

Among the most common rocket nozzle configurations, the bell nozzle offers a strong compromise between performance and compactness. A purely conical nozzle is simple to manufacture and analyze, but its constant wall angle produces larger divergence losses for a given length. A bell nozzle replaces most of the conical diverging section with a curved contour, allowing strong initial expansion immediately downstream of the throat and a smaller exit angle at the nozzle exit. This improves axial momentum alignment while maintaining a shorter nozzle than an equivalent low-angle conical design.

The simulator described in this work was created to generate and analyze single bell nozzle geometries for hybrid rocket engines, with emphasis on a paraffin wax and nitrous oxide propulsion system. The original work mixed implementation details, code excerpts, theoretical derivations, geometry construction and results in the same sections. This revised manuscript separates the material into four layers: physical theory, geometric formulation, numerical implementation logic, and results/optimization interpretation.

The simulator is intended as a preliminary engineering tool rather than a final qualification model. Its value lies in the explicit connection between geometry, thermochemical properties, quasi-one-dimensional flow, viscous-loss closures, visualization, CAD export and performance trends. The current implementation integrates these models with a genetic algorithm so that the expansion ratio and divergent-contour parameters can be optimized automatically at a specified operating point.

2 Objectives and Scope

The simulator has two main purposes. The first is to provide a fast preliminary design tool for a single bell nozzle, from pre-sized propulsion inputs to a usable geometric contour. The second is to optimize the resulting parametric geometry automatically through a genetic algorithm coupled to the same physical model used by the interactive simulator.

The present formulation focuses on:

1. geometric definition of a single bell nozzle contour;
2. use of a throat radius supplied by the engine pre-sizing stage;
3. calculation of exit radius from a prescribed expansion ratio;
4. estimation of an ideal expansion ratio for a chosen operating altitude or ambient pressure;
5. computation of ideal isentropic flow properties along the nozzle;
6. three-dimensional visualization by revolution of the two-dimensional contour;
7. coordinate export for CAD modeling and manufacturing preparation;
8. optimization of expansion ratio, bell-length fraction and initial divergent-wall angle using an effective ambient thrust coefficient.

The simulator is not intended to replace detailed CFD, method-of-characteristics nozzle design, finite-rate chemistry, conjugate thermal analysis, structural analysis or experimental validation. Instead, it provides a fast, interpretable and modifiable preliminary model. Its strongest value is not absolute prediction accuracy, but the ability to connect geometric parameters to performance trends in a controlled and computationally inexpensive way.

3 Baseline Quasi-One-Dimensional Inviscid Nozzle Flow Model

3.1 Thermodynamic closure and governing relations

3.1.1 Governing assumptions

The baseline analysis is based on the classical quasi-one-dimensional nozzle model. The following assumptions are used in the ideal formulation:

- the flow is steady;
- the flow is adiabatic;
- the flow is inviscid in the core region;
- the nozzle flow is shock-free;
- the gas expansion is treated as isentropic;
- the flow is quasi-one-dimensional, meaning that properties vary mainly along the nozzle axis;
- the gas is approximated as a perfect gas with an effective specific heat ratio γ ;
- sonic conditions occur at the throat for a choked nozzle;
- the throat area is equal to the critical area, $A_t = A^*$;
- the subsonic branch of the area–Mach relation is used upstream of the throat;
- the supersonic branch is used downstream of the throat.

These assumptions are appropriate for preliminary design. They are not sufficient for final qualification of a rocket nozzle because they neglect real-gas effects, finite-rate chemistry, wall heat transfer, boundary-layer separation, shocks, nonuniform inlet conditions, two-phase particles or droplets, and multidimensional wave structures.

3.1.2 Thermochemical closure using NASA CEA/RocketCEA

The thermochemical properties used by the simulator are obtained from NASA CEA through RocketCEA. For each combination of chamber pressure, mixture ratio and expansion ratio, RocketCEA can provide useful quantities such as combustion temperature, effective ratio of specific heats, molar mass of combustion products, characteristic velocity, thrust coefficient, exit Mach number, exit temperature, exit pressure and ideal specific impulse.

The classical relations below are written for a locally constant effective γ . In the implemented axial reconstruction, RocketCEA chamber, throat and exit values are interpolated along the contour and the local interpolated value is used in the area–Mach and stagnation-to-static relations. This remains an engineering property closure rather than a finite-rate or station-by-station equilibrium chemistry solution.

For a hybrid engine using nitrous oxide and paraffin-based fuel, RocketCEA also allows the user to define a custom fuel composition and evaluate equilibrium combustion properties for the chosen oxidizer-to-fuel ratio.

3.1.3 Choked mass-flow constraint and external throat pre-sizing

For a choked nozzle, the mass flow rate through the throat is governed by the stagnation conditions upstream of the throat and by the throat area. Assuming isentropic perfect-gas flow, the choked mass flow rate is

$$\dot{m} = A_t P_0 \sqrt{\frac{\gamma}{RT_0}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}. \quad (1)$$

Solving for throat area gives

$$A_t = \frac{\dot{m}}{P_0 \sqrt{\frac{\gamma}{RT_0}} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}}, \quad (2)$$

with

$$r_t = \sqrt{\frac{A_t}{\pi}}. \quad (3)$$

This relation links the propulsion operating point to the geometry. For a fixed chamber pressure and combustion temperature, increasing mass flow requires a larger throat area. For a fixed mass flow rate, increasing chamber pressure allows the same mass flow through a smaller throat. In the present software, however, this calculation belongs to the upstream engine pre-sizing step: r_t is supplied by the user and is neither recalculated nor optimized by the nozzle genetic algorithm.

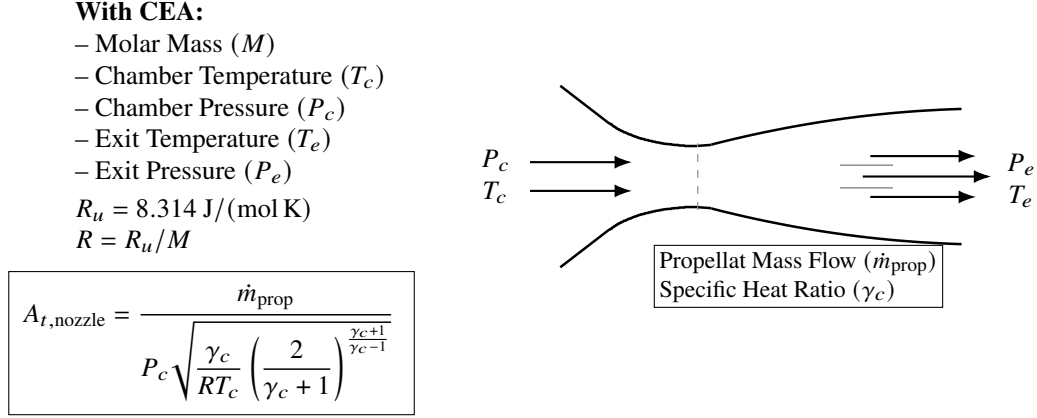


Figure 1: Nozzle flow stations used for external throat pre-sizing and ideal one-dimensional analysis.

3.1.4 Thrust balance and ideal expansion criterion

The thrust produced by a rocket nozzle can be approximated by

$$F = \dot{m}V_e + (p_e - p_a)A_e, \quad (4)$$

where V_e is the exit velocity, p_e is the nozzle exit pressure, p_a is the ambient pressure and A_e is the exit area.

The first term is the momentum thrust. The second term is the pressure thrust. For a fixed chamber condition and mass flow rate, the ideal expansion condition at a given altitude is approximately

$$p_e = p_a. \quad (5)$$

When $p_e > p_a$, the nozzle is underexpanded and the gases still have remaining expansion potential at the exit. When $p_e < p_a$, the nozzle is overexpanded, which can lead to compression waves, shocks, separation risk and loss of performance depending on the operating regime.

Because ambient pressure varies during flight, a fixed-geometry nozzle cannot be perfectly expanded at all altitudes. For low-altitude or short-duration flights, the simulator may use a representative ambient pressure, such as sea-level pressure or a trajectory-averaged ambient pressure. For a more complete flight analysis, the expansion ratio should be evaluated over the expected altitude range and the selected value should minimize a trajectory-weighted performance penalty.

3.1.5 Expansion ratio and exit-plane radius

The expansion ratio is defined as

$$\varepsilon = \frac{A_e}{A_t}. \quad (6)$$

For circular sections,

$$\varepsilon = \frac{\pi r_e^2}{\pi r_t^2} = \left(\frac{r_e}{r_t} \right)^2, \quad (7)$$

therefore

$$r_e = \sqrt{\varepsilon} r_t. \quad (8)$$

The expansion ratio is one of the most important nozzle design variables. Increasing ε generally increases exit Mach number and ideal exhaust velocity, but also increases nozzle length, exit area, structural mass and overexpansion risk at low altitude.

3.1.6 Propellant mass-flow closure from mixture ratio

Using the conventional definition

$$O/F = \frac{\dot{m}_{ox}}{\dot{m}_{fuel}}, \quad (9)$$

the fuel mass flow rate is

$$\dot{m}_{fuel} = \frac{\dot{m}_{ox}}{O/F}, \quad (10)$$

and the total propellant mass flow rate is

$$\dot{m}_{prop} = \dot{m}_{ox} + \dot{m}_{fuel} = \dot{m}_{ox} \left(1 + \frac{1}{O/F} \right). \quad (11)$$

This convention must be used consistently. Reversing the definition of mixture ratio changes the mass-flow calculation and therefore the throat sizing.

3.2 Axial flow-field reconstruction

3.2.1 Stagnation-to-static isentropic relations

For one-dimensional, steady, adiabatic and inviscid flow of a perfect gas, the stagnation-to-static relations are

$$\frac{T}{T_0} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-1}, \quad (12)$$

$$\frac{p}{p_0} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-\gamma/(\gamma-1)}, \quad (13)$$

$$\frac{\rho}{\rho_0} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-1/(\gamma-1)}. \quad (14)$$

Once the Mach number distribution $M(x)$ is known, these relations provide the axial distributions of temperature, pressure and density.

3.2.2 Area–Mach compatibility relation

The Mach number is implicitly related to the local cross-sectional area through the area–Mach relation:

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}. \quad (15)$$

For a choked nozzle, $A^* = A_t$. Therefore, at each axial station,

$$\frac{A(x)}{A_t} = \frac{\pi r(x)^2}{\pi r_t^2} = \left(\frac{r(x)}{r_t} \right)^2. \quad (16)$$

Equation (15) has two possible Mach solutions for $A/A^* > 1$: a subsonic solution and a supersonic solution. The correct branch is selected from the location relative to the throat. Upstream of the throat, the subsonic solution is used. Downstream of the throat, the supersonic solution is used. At the throat, $M = 1$.

3.2.3 Branch-resolved numerical solution of the Mach number

Because Eq. (15) cannot generally be inverted explicitly for M , the simulator determines $M(x)$ by solving

$$f(M) = \frac{A}{A^*} - \frac{1}{M} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}} = 0. \quad (17)$$

The implementation uses Brent's bracketed root method, combining the robustness of a bracketing method with faster interpolation steps when the residual permits them. The numerical intervals are

$$M \in (0, 1) \quad \text{for the subsonic branch,} \quad (18)$$

$$M \in (1, M_{max}) \quad \text{for the supersonic branch.} \quad (19)$$

Once $M(x)$ is obtained, the local speed of sound and axial velocity are

$$a(x) = \sqrt{\gamma RT(x)}, \quad (20)$$

$$u(x) = M(x)a(x). \quad (21)$$

3.2.4 Axial reconstruction of Mach, static-pressure and static-temperature curves

The axial property curves couple the piecewise radius function to the quasi-one-dimensional isentropic model. The process starts from the discretized nozzle contour,

$$\{x_i, r_i\}_{i=0}^N, \quad (22)$$

where each station has the local cross-sectional area

$$A_i = \pi r_i^2. \quad (23)$$

Because the throat is assumed choked, the critical area is

$$A^* = A_t = \pi r_t^2. \quad (24)$$

Therefore, the local area ratio used in the area–Mach equation is

$$\Lambda_i = \frac{A_i}{A_t} = \left(\frac{r_i}{r_t} \right)^2. \quad (25)$$

For each station, the Mach number is obtained by solving the residual form of the area–Mach equation,

$$F(M_i, \Lambda_i) = \frac{1}{M_i} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_i^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}} - \Lambda_i = 0. \quad (26)$$

The solution branch is selected from the geometric station location:

$$0 < M_i < 1 \quad \text{for } x_i < x_t, \quad (27)$$

$$M_i = 1 \quad \text{for } x_i = x_t, \quad (28)$$

$$M_i > 1 \quad \text{for } x_i > x_t. \quad (29)$$

This branch selection is essential because the area–Mach relation is double-valued for $A/A^* > 1$. The same area ratio may correspond either to a subsonic state upstream of the throat or to a supersonic state downstream of it.

After the Mach number has been obtained at each station, the static-temperature curve is generated directly from the stagnation-to-static relation,

$$T_i = T(x_i) = \frac{T_0}{1 + \frac{\gamma-1}{2} M_i^2}. \quad (30)$$

Similarly, the static-pressure curve is obtained from

$$p_i = p(x_i) = p_0 \left(1 + \frac{\gamma - 1}{2} M_i^2 \right)^{-\gamma/(\gamma-1)}. \quad (31)$$

The density curve may also be reconstructed as

$$\rho_i = \rho(x_i) = \rho_0 \left(1 + \frac{\gamma - 1}{2} M_i^2 \right)^{-1/(\gamma-1)}. \quad (32)$$

The simulator therefore stores the axial flow-field dataset as

$$\mathcal{D}_{flow} = \{x_i, A_i, \Lambda_i, M_i, T_i, p_i, \rho_i, u_i\}_{i=0}^N, \quad (33)$$

where the velocity is obtained from

$$u_i = M_i \sqrt{\gamma R T_i}. \quad (34)$$

The Mach, pressure and temperature curves are then plotted as functions of the translated axial coordinate $\bar{x}$. This means that the plotted distributions are not read directly from CEA along the nozzle. CEA/RocketCEA provides the thermochemical closure and reference stagnation properties, while the spatial evolution along the generated contour is reconstructed from $r(x)$, the area–Mach relation and the isentropic equations.

In the converging subsonic region, reducing area accelerates the flow and increases Mach number toward unity. At the throat, the nozzle becomes choked and $M = 1$. In the diverging supersonic region, increasing area further accelerates the flow and produces a supersonic Mach number increase.

The temperature and pressure decrease throughout the expansion according to Eqs. (12) and (13). Therefore, the simulator can generate axial distributions of Mach number, pressure and temperature from the geometric contour alone, once the chamber stagnation state and thermodynamic model have been selected.

4 Gas-Side Thermal Model

4.1 Model scope

The implemented thermal module reports the local adiabatic-wall temperature, $T_{aw}(x)$, and a Bartz gas-side heat-transfer coefficient, $h_g(x)$. The coefficient is retained as a diagnostic indicator of where gas-side convective coupling would be strongest; it is not combined with an independently prescribed wall temperature. Thermophysical properties are supplied by RocketCEA and mapped onto the quasi-one-dimensional flow reconstruction.

No conjugate wall-conduction, coolant or ablative model is solved. The adopted gas-side boundary condition is therefore adiabatic,

$$\boxed{T_w(x) = T_{aw}(x)}, \quad \boxed{q''_w(x) = 0}. \quad (35)$$

Consequently, the thermal output must not be interpreted as a prediction of material temperature or cooling-system heat load.

4.2 Adiabatic wall temperature and recovery factor

For the fully turbulent closure, the local recovery factor is derived from the local edge Prandtl number,

$$r_f(x) = Pr_e(x)^{1/3}. \quad (36)$$

The edge Prandtl number is obtained by interpolation of the RocketCEA chamber, throat and exit transport values; it is not a user input. The adiabatic-wall temperature is then

$$\boxed{T_{aw}(x) = T_e(x) \left[1 + r_f(x) \frac{\gamma_e(x) - 1}{2} M_e(x)^2 \right]}. \quad (37)$$

This recovery-temperature condition is also used by the compressible boundary-layer closures. The associated adiabatic Walz temperature–velocity relation is developed in Sections 5 and 6 [8].

4.3 Diagnostic Bartz heat-transfer coefficient

The gas-side convective heat-transfer coefficient is estimated using the simplified Bartz correlation [9],

$$h_g(x) = \frac{0.026 \mu_t^{0.2} c_{p,t}}{D_t^{0.2} Pr_t^{0.6}} \left(\frac{p_c}{c^*} \right)^{0.8} \left(\frac{D_t}{R_{c,t}} \right)^{0.1} \left(\frac{A_t}{A(x)} \right)^{0.9} \sigma(x), \quad (38)$$

where $D_t = 2r_t$, $R_{c,t} = 0.4r_t$ is the divergent throat-arc radius, and μ_t , $c_{p,t}$ and Pr_t are evaluated at the throat. The local area ratio follows directly from the generated contour,

$$\frac{A_t}{A(x)} = \left(\frac{r_t}{r(x)} \right)^2. \quad (39)$$

The Bartz correction factor is evaluated consistently with the adiabatic wall,

$$\begin{aligned} \sigma(x) = & \left[\frac{1}{2} \frac{T_{aw}(x)}{T_0} \left(1 + \frac{\gamma_e(x) - 1}{2} M_e(x)^2 \right) + \frac{1}{2} \right]^{-0.68} \\ & \times \left(1 + \frac{\gamma_e(x) - 1}{2} M_e(x)^2 \right)^{-0.12}. \end{aligned} \quad (40)$$

Although h_g generally peaks near the throat and remains useful for comparing the spatial intensity of convective coupling, the present adiabatic boundary condition gives

$$q''_g(x) = h_g(x) [T_{aw}(x) - T_w(x)] = 0. \quad (41)$$

4.4 Interpretation of the adiabatic result

Setting $T_w = T_{aw}$ closes the gas-side boundary-layer problem without inventing a geometry-independent solid temperature. It does not imply that a real nozzle is thermally insulated, nor does it predict that the physical wall reaches the reported temperature. A non-zero heat flux requires an additional thermal boundary condition and a model that determines T_w , for example conduction through the nozzle wall coupled to regenerative, film or ablative cooling.

The current model omits wall thickness and material properties, transient energy storage, radiation, coolant-channel geometry and coolant-side heat transfer. It therefore supports aerodynamic preliminary design only. Thermal sizing and material selection require a conjugate thermal model or experimental data after the geometry has been defined.

5 Integral Compressible Boundary-Layer Model for Quick Screening

The internal flow of a rocket nozzle is characterized by high velocities, strong streamwise acceleration, and large variations in pressure, temperature, density, and viscosity. Furthermore, the generally high Reynolds numbers make the near-wall flow predominantly turbulent. Consequently, an incompressible flat-plate correlation is not sufficient to represent the boundary-layer development inside the nozzle.

Resolving the complete viscous flow would require a two-dimensional or three-dimensional CFD model. The simulator therefore provides two reduced-order closures. This section describes the inexpensive integral model used by the *Quick screening* optimization mode. The nozzle core remains quasi-one-dimensional, while the integrated mass-flow and momentum-flow deficits produced by the unresolved boundary layer are transported along the nozzle wall. Final evaluation uses the BLIMP-inspired profile marcher developed in Section 6.

The general conservative transport equation can be written as

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}^c = \nabla \cdot \mathbf{F}^d + \mathbf{S}, \quad (42)$$

where $\mathbf{U}$ is the vector of conserved variables, $\mathbf{F}^c$ is the convective-flux tensor, $\mathbf{F}^d$ is the diffusive-flux tensor, and $\mathbf{S}$ contains the remaining source terms.

For conservation of linear momentum, the transported conserved variable is

$$\mathbf{U}_m = \rho \mathbf{u}. \quad (43)$$

Neglecting body forces, the local momentum equation is

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau}, \quad (44)$$

where p is the static pressure and $\boldsymbol{\tau}$ is the total shear-stress tensor. For turbulent flow, $\boldsymbol{\tau}$ represents the combined molecular and modeled turbulent transport of momentum.

5.1 Coordinate System and Quasi-One-Dimensional Core

The boundary layer is described using the wall-fitted coordinates s and n . The coordinate s is measured tangentially along the nozzle wall, whereas n is measured locally normal to the wall and towards the nozzle core.

The nozzle contour is originally defined in axial coordinates by

$$r = r(x). \quad (45)$$

The differential wall distance is

$$ds = \sqrt{dx^2 + dr^2} = \sqrt{1 + \left(\frac{dr}{dx}\right)^2} dx. \quad (46)$$

Consequently,

$$s(x) = \int_{x_0}^x \sqrt{1 + \left(\frac{dr}{d\xi}\right)^2} d\xi. \quad (47)$$

On the discrete nozzle grid,

$$s_i = s_{i-1} + \sqrt{(x_i - x_{i-1})^2 + (r_i - r_{i-1})^2}. \quad (48)$$

The local wall radius is therefore

$$R_w(s) = r(x(s)), \quad (49)$$

and the corresponding wetted perimeter is

$$P_w(s) = 2\pi R_w(s) = 2\pi r(x(s)). \quad (50)$$

The quasi-one-dimensional inviscid model provides the core-flow properties

$$M(x), \quad T(x), \quad p(x), \quad \rho(x), \quad \gamma(x). \quad (51)$$

These properties are mapped onto the wall coordinate through the one-to-one correspondence $x_i \leftrightarrow s_i$:

$$\{M(x_i), T(x_i), p(x_i), \rho(x_i), \gamma(x_i)\} \longrightarrow \{M_e(s_i), T_e(s_i), p_e(s_i), \rho_e(s_i), \gamma_e(s_i)\}. \quad (52)$$

The subscript e denotes a property evaluated at the outer edge of the boundary layer. In particular, the edge velocity is approximated by the local quasi-one-dimensional core velocity:

$$U_e(s_i) \approx U_{Q1D}(x_i) = M(x_i) \sqrt{\gamma(x_i) R_g(x_i) T(x_i)}, \quad x_i \leftrightarrow s_i, \quad (53)$$

where R_g is the specific gas constant and must not be confused with the wall radius R_w .

The edge velocity $U_e(s)$ is therefore a one-dimensional quantity. It is not a velocity profile across the boundary layer.

Figure 2 illustrates the relation between the quasi-one-dimensional core and the local wall-fitted coordinate system.

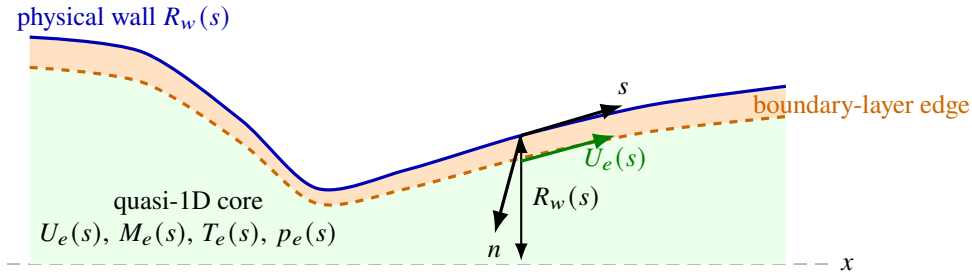


Figure 2: Quasi-one-dimensional core and wall-fitted coordinates used by the integral boundary-layer model.

No additional resolved spatial dimension is introduced by the mapping $x \leftrightarrow s$. The inviscid core remains quasi-one-dimensional, and the transported integral boundary-layer variables are also functions of the single coordinate s .

5.2 Velocity Components Inside the Boundary Layer

The actual local velocity field inside the boundary layer contains tangential and wall-normal components:

$$\mathbf{u}(s, n) = u_s(s, n)\mathbf{e}_s + u_n(s, n)\mathbf{e}_n. \quad (54)$$

Here,

- $u_s(s, n)$ is the velocity component tangent to the wall;
- $u_n(s, n)$ is the velocity component normal to the wall;
- $U_e(s)$ is the one-dimensional tangential velocity at the boundary-layer edge.

At a stationary, impermeable, no-slip wall,

$$u_s(s, 0) = 0, \quad u_n(s, 0) = 0. \quad (55)$$

At the outer edge of the boundary layer,

$$u_s(s, \delta(s)) \simeq U_e(s), \quad \tau_{sn}(s, \delta(s)) \simeq 0. \quad (56)$$

The wall-normal component is zero at the impermeable wall but is generally nonzero inside the boundary layer:

$$u_n(s, n) \neq 0, \quad 0 < n \leq \delta(s). \quad (57)$$

Its magnitude is normally much smaller than the tangential velocity:

$$|u_n| \ll |u_s|. \quad (58)$$

Nevertheless, the normal convection term cannot be discarded because the normal velocity gradient is large inside a thin boundary layer. In general,

$$u_n \frac{\partial u_s}{\partial n} \quad (59)$$

can be of the same asymptotic order as

$$u_s \frac{\partial u_s}{\partial s}. \quad (60)$$

The present model does not explicitly resolve $u_s(s, n)$, $u_n(s, n)$, $T(s, n)$, or $\rho(s, n)$. These two-dimensional fields appear in the theoretical definitions and derivation, but their integrated effects are represented by one-dimensional boundary-layer variables.

5.3 Physical and Integral Boundary-Layer Thicknesses

Three different boundary-layer thicknesses are employed:

$$\boxed{\delta(s), \quad \delta^*(s), \quad \theta(s)}. \quad (61)$$

Although all three quantities have dimensions of length, they represent different physical effects.

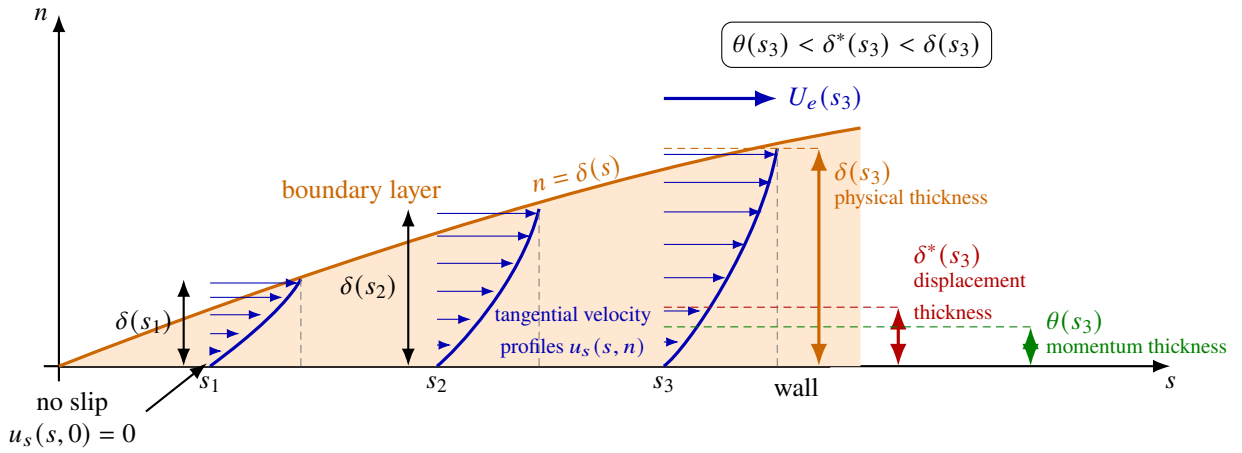


Figure 3: Development of the tangential velocity profile along the nozzle wall and comparison of the conventional boundary-layer thickness δ , displacement thickness δ^* , and momentum thickness θ at station s_3 . The coordinate n is normal to the wall, whereas s follows the wall in the streamwise direction. The displayed relative magnitudes of θ and δ^* are schematic and are not drawn to scale.

5.3.1 Representative physical thickness

The velocity approaches U_e asymptotically, and therefore the boundary layer does not possess an exact physical edge. A common engineering convention is to identify its thickness with the wall-normal location at which

$$\frac{u_s}{U_e} \simeq 0.99. \quad (62)$$

The quantity $\delta(s)$ is consequently interpreted as a representative physical thickness of the velocity-gradient region. In the profile closure introduced below, it is treated as the outer scale of the assumed velocity profile and is approximately identified with the conventional δ_{99} .

5.3.2 Compressible displacement thickness

Inside the boundary layer, the local mass flux ρu_s is smaller than the mass flux $\rho_e U_e$ that would exist if the inviscid edge state extended uniformly to the physical wall.

The mass-flow deficit per unit wetted perimeter is

$$\dot{m}'_{\text{def}}(s) = \int_0^{\delta(s)} [\rho_e(s)U_e(s) - \rho(s,n)u_s(s,n)] dn. \quad (63)$$

Factoring out the edge mass flux gives

$$\dot{m}'_{\text{def}}(s) = \rho_e(s)U_e(s) \int_0^{\delta(s)} \left[1 - \frac{\rho(s,n)u_s(s,n)}{\rho_e(s)U_e(s)} \right] dn. \quad (64)$$

The compressible displacement thickness is therefore defined as

$$\delta^*(s) = \int_0^{\delta(s)} \left[1 - \frac{\rho(s,n)u_s(s,n)}{\rho_e(s)U_e(s)} \right] dn. \quad (65)$$

It follows that

$$\dot{m}'_{\text{def}}(s) = \rho_e(s)U_e(s)\delta^*(s). \quad (66)$$

The displacement thickness is the thickness of a hypothetical region carrying zero mass flow that would produce the same mass-flow deficit as the actual boundary layer.

Equivalently, the inviscid core behaves approximately as if the solid wall were displaced towards the nozzle axis by δ^* . Under the thin-layer approximation, the effective core radius is

$$R_{\text{eff}}(s) = R_w(s) - \delta^*(s). \quad (67)$$

The displacement thickness is not the physical location of the boundary-layer edge. Generally,

$$\delta^*(s) < \delta(s). \quad (68)$$

5.3.3 Compressible momentum thickness

The reduced tangential velocity also produces a deficit in streamwise momentum flux. The momentum-flux deficit per unit wetted perimeter is

$$D'_m(s) = \int_0^{\delta(s)} \rho(s,n)u_s(s,n) [U_e(s) - u_s(s,n)] dn. \quad (69)$$

Introducing the edge quantities gives

$$D'_m(s) = \rho_e(s)U_e^2(s) \int_0^{\delta(s)} \frac{\rho(s,n)u_s(s,n)}{\rho_e(s)U_e(s)} \left[1 - \frac{u_s(s,n)}{U_e(s)} \right] dn. \quad (70)$$

The compressible momentum thickness is defined as

$$\theta(s) = \int_0^{\delta(s)} \frac{\rho(s,n)u_s(s,n)}{\rho_e(s)U_e(s)} \left[1 - \frac{u_s(s,n)}{U_e(s)} \right] dn. \quad (71)$$

Consequently,

$$\boxed{D'_m(s) = \rho_e(s)U_e^2(s)\theta(s)}, \quad [D'_m] = \text{N m}^{-1}. \quad (72)$$

The momentum thickness is the thickness of a hypothetical region producing the same momentum-flux deficit as the actual velocity and density profiles. It is not itself momentum and does not represent a physical interface inside the boundary layer.

For the complete circumference of an axisymmetric nozzle,

$$\boxed{D_m(s) = P_w(s)\rho_e(s)U_e^2(s)\theta(s)} \quad (73)$$

or, using $P_w = 2\pi R_w$,

$$\boxed{D_m(s) = 2\pi R_w(s)\rho_e(s)U_e^2(s)\theta(s)}, \quad [D_m] = \text{N}. \quad (74)$$

The three thicknesses can therefore be summarized as

$$\delta : \quad \text{representative physical extent of the velocity-gradient region}, \quad (75)$$

$$\delta^* : \quad \text{equivalent thickness associated with mass-flow deficit}, \quad (76)$$

$$\theta : \quad \text{equivalent thickness associated with momentum-flux deficit}. \quad (77)$$

For a conventional attached turbulent boundary layer, the typical ordering is

$$\boxed{\theta(s) < \delta^*(s) < \delta(s)}. \quad (78)$$

The ratio

$$\boxed{H(s) = \frac{\delta^*(s)}{\theta(s)}} \quad (79)$$

is the compressible boundary-layer shape factor. It provides an integral measure of the shapes of the velocity and density profiles.

Figure 4 illustrates the different meanings of δ , δ^* , and θ .

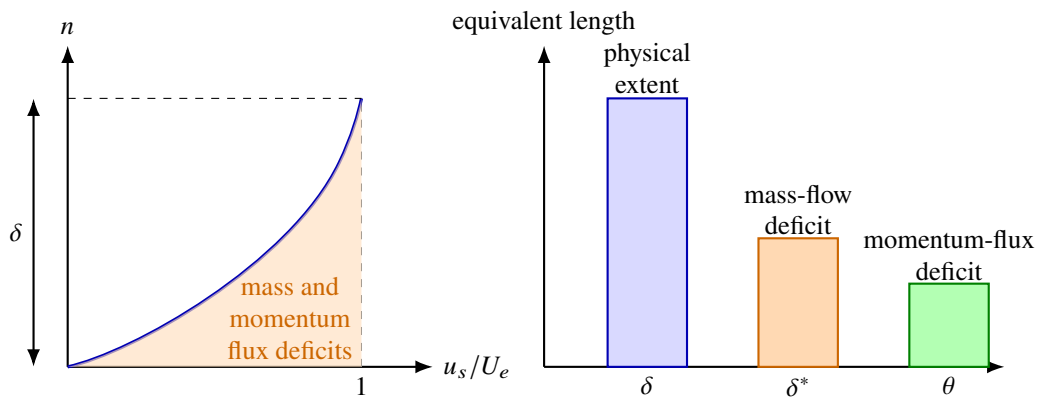


Figure 4: Physical interpretation of the representative boundary-layer thickness δ , displacement thickness δ^* , and momentum thickness θ . The relative dimensions are schematic and not to scale.

5.4 Normalized Profile Closure

The definitions of δ^* and θ depend on the wall-normal profiles

$$u_s(s, n), \quad T(s, n), \quad \rho(s, n), \quad (80)$$

and on the physical thickness $\delta(s)$. These quantities are not provided by the quasi-one-dimensional core model.

To evaluate the shape factor without resolving the complete two-dimensional boundary layer, normalized velocity, temperature, and density profiles are introduced.

5.4.1 Normalized wall-normal coordinate

The dimensionless coordinate

$$\boxed{\eta = \frac{n}{\delta(s)}}, \quad 0 \leq \eta \leq 1, \quad (81)$$

maps the boundary layer at every streamwise station onto the same normalized interval.

Thus,

$$\eta = 0 \iff n = 0, \quad \text{wall}, \quad (82)$$

$$\eta = 1 \iff n = \delta(s), \quad \text{representative boundary-layer edge.} \quad (83)$$

The dimensional and normalized coordinates are related by

$$n = \eta \delta(s), \quad dn = \delta(s) d\eta. \quad (84)$$

The coordinate η is not an additional resolved spatial dimension. It is an auxiliary quadrature coordinate used to describe the assumed shapes of the unresolved profiles.

The normalization is useful because $\delta(s)$ is not initially known. Although the dimensional values of δ^* and θ cannot yet be calculated from their definitions, their ratios to δ can be calculated using normalized profiles.

5.4.2 One-seventh-power velocity profile

The tangential velocity profile is approximated using the empirical turbulent one-seventh-power law:

$$\boxed{f(\eta) \equiv \frac{u_s(s, n)}{U_e(s)} = \eta^{1/7}}. \quad (85)$$

The profile satisfies the idealized boundary conditions

$$f(0) = 0, \quad f(1) = 1. \quad (86)$$

The first condition represents no slip at the wall. The second condition matches the assumed profile to the edge velocity. The conventional 99% definition of δ and the condition $f(1) = 1$ are treated as equivalent engineering representations of the boundary-layer edge.

The one-seventh-power law prescribes only the normalized shape of the velocity profile. It does not determine the dimensional thickness $\delta(s)$.

Figure 5 illustrates the normalized coordinate and the assumed velocity and temperature profiles.

5.4.3 Recovery temperature and adiabatic wall

The static edge temperature $T_e(s)$ does not represent the complete temperature distribution across the boundary layer. In compressible flow, part of the kinetic energy is recovered as internal energy near the wall.

The corresponding recovery temperature is

$$\boxed{T_{\text{rec}}(s) = T_e(s) \left[1 + r_{\text{rec}}(s) \frac{\gamma_e(s) - 1}{2} M_e^2(s) \right]}, \quad (87)$$

where r_{rec} is the recovery factor. For a turbulent boundary layer,

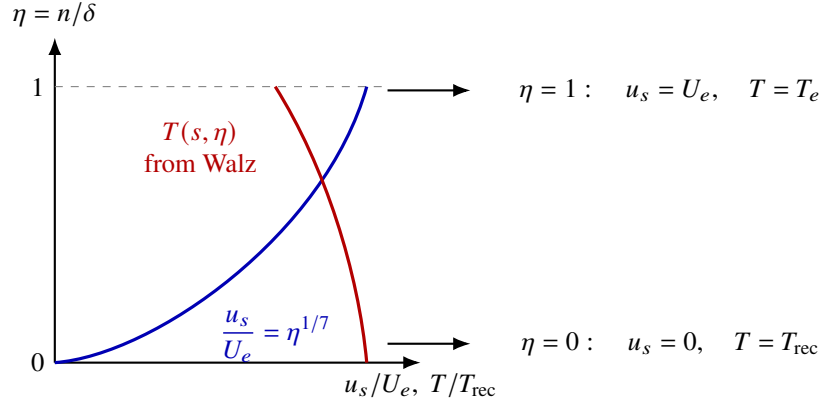


Figure 5: Normalized wall-normal coordinate and assumed profiles for an adiabatic compressible turbulent boundary layer. The diagram is schematic.

$$\boxed{r_{\text{rec}}(s) \simeq Pr_e^{1/3}(s)}. \quad (88)$$

The recovery temperature is the temperature approached by an adiabatic wall under the local edge-flow conditions. It is generally bounded by

$$T_e(s) \leq T_{\text{rec}}(s) \leq T_0(s), \quad (89)$$

where T_0 is the local stagnation temperature.

The present model assumes an adiabatic wall:

$$q_w'' = 0. \quad (90)$$

Consequently,

$$\boxed{T_w(s) = T_{\text{rec}}(s)}. \quad (91)$$

The wall temperature is therefore not an independent prescribed input in the adiabatic boundary-layer closure.

5.4.4 Walz temperature–velocity relation

The density profile required by δ^* and θ depends on the temperature across the boundary layer. Since the present model does not solve a wall-normal energy equation, the temperature profile is approximated using the Walz temperature–velocity relation.

For a general wall temperature, the Walz relation is

$$\boxed{T(s, \eta) = T_w(s) + [T_{\text{rec}}(s) - T_w(s)] f(\eta) - [T_{\text{rec}}(s) - T_e(s)] f^2(\eta)}. \quad (92)$$

There is no single quantity referred to here as the “Walz temperature.” Equation (92) provides the complete approximate temperature profile

$$T = T(s, \eta) \quad (93)$$

as a function of the local normalized velocity $f = u_s/U_e$.

Under the adiabatic-wall condition $T_w = T_{\text{rec}}$, the linear term vanishes and the relation reduces to

$$\boxed{T(s, \eta) = T_{\text{rec}}(s) - [T_{\text{rec}}(s) - T_e(s)] f^2(\eta)}. \quad (94)$$

Using $f(\eta) = \eta^{1/7}$,

$$T(s, \eta) = T_{\text{rec}}(s) - [T_{\text{rec}}(s) - T_e(s)] \eta^{2/7}. \quad (95)$$

The limiting values are

$$T(s, 0) = T_{\text{rec}}(s) = T_w(s), \quad (96)$$

$$T(s, 1) = T_e(s). \quad (97)$$

The relation therefore describes the temperature variation from the recovery temperature at the adiabatic wall to the static edge temperature at the outer boundary of the layer.

5.4.5 Normalized density profile

The thin-boundary-layer approximation implies that the static pressure is approximately uniform in the wall-normal direction:

$$p(s, n) \approx p_e(s), \quad \frac{\partial p}{\partial n} \approx 0. \quad (98)$$

This does not imply that the density is uniform across the boundary layer. Since

$$T = T(s, n), \quad (99)$$

the density also depends on the wall-normal coordinate:

$$\rho = \rho(s, n). \quad (100)$$

Assuming ideal-gas behavior and an approximately uniform specific gas constant across the thin layer,

$$\rho(s, \eta) = \frac{p_e(s)}{R_g(s)T(s, \eta)}, \quad (101)$$

$$\rho_e(s) = \frac{p_e(s)}{R_g(s)T_e(s)}. \quad (102)$$

Dividing these expressions gives the normalized density profile

$$\frac{\rho(s, \eta)}{\rho_e(s)} = \frac{T_e(s)}{T(s, \eta)}. \quad (103)$$

The assumed profile closure can therefore be summarized as

$$\eta \longrightarrow \frac{u_s}{U_e} \longrightarrow T(s, \eta) \longrightarrow \frac{\rho(s, \eta)}{\rho_e(s)}. \quad (104)$$

The one-seventh-power law provides the normalized velocity profile, the Walz relation provides the corresponding temperature profile, and the ideal-gas relation provides the density profile required by the compressible integral thicknesses.

5.4.6 Normalized integral thicknesses

Using

$$n = \eta \delta(s), \quad dn = \delta(s) d\eta, \quad (105)$$

the displacement thickness becomes

$$\delta^*(s) = \delta(s) \int_0^1 \left[1 - \frac{\rho(s, \eta)}{\rho_e(s)} f(\eta) \right] d\eta. \quad (106)$$

The dimensionless displacement-thickness ratio is therefore

$$I_{\delta^*}(s) \equiv \frac{\delta^*(s)}{\delta(s)} = \int_0^1 \left[1 - \frac{\rho(s, \eta)}{\rho_e(s)} f(\eta) \right] d\eta. \quad (107)$$

Similarly, the momentum thickness becomes

$$\theta(s) = \delta(s) \int_0^1 \frac{\rho(s, \eta)}{\rho_e(s)} f(\eta) [1 - f(\eta)] d\eta. \quad (108)$$

The corresponding dimensionless ratio is

$$I_{\theta}(s) \equiv \frac{\theta(s)}{\delta(s)} = \int_0^1 \frac{\rho(s, \eta)}{\rho_e(s)} f(\eta) [1 - f(\eta)] d\eta. \quad (109)$$

At this stage, the dimensional boundary-layer thickness $\delta(s)$ remains unknown. Consequently, these profile integrations do not directly provide δ^* or θ in metres.

However, the unknown thickness cancels in their ratio:

$$H(s) = \frac{\delta^*(s)}{\theta(s)} = \frac{\delta(s) I_{\delta^*}(s)}{\delta(s) I_{\theta}(s)}. \quad (110)$$

Therefore,

$$H(s) = \frac{I_{\delta^*}(s)}{I_{\theta}(s)} \quad (111)$$

or, explicitly,

$$H(s) = \frac{\int_0^1 \left[1 - \frac{\rho}{\rho_e} f \right] d\eta}{\int_0^1 \frac{\rho}{\rho_e} f (1 - f) d\eta}. \quad (112)$$

This cancellation is the principal reason for introducing the normalized coordinate η . The shape factor can be calculated before the dimensional boundary-layer thickness is known.

Numerically, the interval $0 \leq \eta \leq 1$ is divided into a set of quadrature points. At every streamwise station s_i , the model evaluates

$$f(\eta_j), \quad T(s_i, \eta_j), \quad \frac{\rho(s_i, \eta_j)}{\rho_e(s_i)}, \quad (113)$$

and integrates Eqs. (107) and (109) using the trapezoidal rule.

Once $\theta(s)$ has been obtained from the integral momentum equation, the remaining thicknesses can be recovered from

$$\delta^*(s) = H(s) \theta(s) \quad (114)$$

and

$$\delta(s) = \frac{\theta(s)}{I_{\theta}(s)}. \quad (115)$$

5.5 Local Conservative Momentum Equation

For statistically steady flow,

$$\frac{\partial(\rho \mathbf{u})}{\partial t} = 0. \quad (116)$$

The conservative momentum equation therefore reduces to

$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau}. \quad (117)$$

Resolving this equation in the direction tangent to the wall gives

$$\frac{\partial(\rho u_s^2)}{\partial s} + \frac{\partial(\rho u_n u_s)}{\partial n} = -\frac{\partial p}{\partial s} + \frac{\partial \tau_{ss}}{\partial s} + \frac{\partial \tau_{sn}}{\partial n}. \quad (118)$$

The first two terms represent the tangential and normal convective transport of tangential momentum. The last two terms represent the tangential and normal diffusive transport of tangential momentum.

For a thin boundary layer,

$$\delta \ll L, \quad \frac{\partial}{\partial n} \gg \frac{\partial}{\partial s}. \quad (119)$$

Consequently, the streamwise diffusion of tangential momentum is much smaller than its wall-normal diffusion:

$$\frac{\partial \tau_{ss}}{\partial s} \ll \frac{\partial \tau_{sn}}{\partial n}. \quad (120)$$

Using also $p(s, n) \simeq p_e(s)$, the local tangential momentum equation reduces to

$$\boxed{\frac{\partial(\rho u_s^2)}{\partial s} + \frac{\partial(\rho u_n u_s)}{\partial n} = -\frac{dp_e}{ds} + \frac{\partial \tau_{sn}}{\partial n}}. \quad (121)$$

The corresponding steady continuity equation is

$$\nabla \cdot (\rho \mathbf{u}) = 0. \quad (122)$$

For a thin axisymmetric boundary layer, it can be expressed approximately in perimeter-weighted form as

$$\boxed{\frac{\partial}{\partial s} [P_w(s) \rho u_s] + \frac{\partial}{\partial n} [P_w(s) \rho u_n] = 0}. \quad (123)$$

The normal velocity u_n is required by local mass conservation. It is zero at the impermeable wall but is generally nonzero inside the boundary layer. The integral model does not solve $u_n(s, n)$ explicitly. Instead, continuity is combined with the tangential momentum equation before integration in the normal direction.

5.6 Integral Momentum Equation

The inviscid edge flow satisfies the streamwise Euler equation

$$\boxed{\rho_e(s) U_e(s) \frac{dU_e}{ds} = -\frac{dp_e}{ds}}. \quad (124)$$

Equations (121), (123), and (124) are combined and integrated from the wall, $n = 0$, to the boundary-layer edge, $n = \delta(s)$.

During this integration, the normal convective transport containing u_n reduces to boundary contributions. At the impermeable wall,

$$u_n(s, 0) = 0, \quad (125)$$

whereas the momentum-deficit integrand vanishes at the boundary-layer edge because

$$u_s(s, \delta(s)) \simeq U_e(s). \quad (126)$$

The wall-normal diffusion term becomes

$$\int_0^{\delta(s)} \frac{\partial \tau_{sn}}{\partial n} dn = \tau_{sn}(s, \delta(s)) - \tau_{sn}(s, 0). \quad (127)$$

Since the shear stress is negligible at the boundary-layer edge,

$$\tau_{sn}(s, \delta(s)) \simeq 0, \quad (128)$$

the integrated normal diffusion is represented by the wall shear stress. Defining its positive magnitude as

$$\tau_w(s) = |\tau_{sn}(s, 0)|, \quad (129)$$

the axisymmetric compressible integral momentum equation becomes

$$\boxed{\frac{d}{ds} [P_w(s) \rho_e(s) U_e^2(s) \theta(s)] + P_w(s) \rho_e(s) U_e(s) \frac{dU_e}{ds} \delta^*(s) = P_w(s) \tau_w(s)}. \quad (130)$$

Since

$$D_m(s) = P_w(s) \rho_e(s) U_e^2(s) \theta(s), \quad (131)$$

the integral equation can also be written as

$$\boxed{\frac{dD_m}{ds} = P_w \tau_w - P_w \rho_e U_e \frac{dU_e}{ds} \delta^*}. \quad (132)$$

This equation shows that the transported integral quantity is the total momentum-flux deficit D_m , rather than θ alone. The momentum thickness is the equivalent length used to express this deficit through

$$D_m = P_w \rho_e U_e^2 \theta. \quad (133)$$

Defining the skin-friction coefficient as

$$\boxed{C_f(s) = \frac{2\tau_w(s)}{\rho_e(s) U_e^2(s)}} \quad (134)$$

and using

$$H(s) = \frac{\delta^*(s)}{\theta(s)}, \quad (135)$$

Eq. (130) can be expanded into an evolution equation for the momentum thickness:

$$\boxed{\frac{d\theta}{ds} = \frac{C_f}{2} - \theta \left[\frac{d \ln P_w}{ds} + \frac{d \ln \rho_e}{ds} + (2 + H) \frac{d \ln U_e}{ds} \right]}. \quad (136)$$

Since $P_w = 2\pi R_w$,

$$\frac{d \ln P_w}{ds} = \frac{d \ln R_w}{ds}, \quad (137)$$

and the equation may equivalently be written as

$$\boxed{\frac{d\theta}{ds} = \frac{C_f}{2} - \theta \left[\frac{d \ln R_w}{ds} + \frac{d \ln \rho_e}{ds} + (2 + H) \frac{d \ln U_e}{ds} \right]}. \quad (138)$$

The terms in this equation represent, respectively,

- production of momentum thickness by wall shear, $C_f/2$;
- the effect of the changing axisymmetric wetted perimeter;
- the effect of the streamwise variation in edge density;
- the effect of core-flow acceleration and pressure gradient.

The integral model therefore transports the one-dimensional state $\theta(s)$, provided that closure relations are supplied for $C_f(s)$ and $H(s)$. The profile closure supplies $H(s)$, while the wall-shear closure supplies $C_f(s)$. After integrating $\theta(s)$, the displacement and representative physical thicknesses are recovered from

$$\delta^*(s) = H(s)\theta(s) \quad (139)$$

and

$$\delta(s) = \frac{\theta(s)}{I_\theta(s)}. \quad (140)$$

5.7 Skin Friction Closure Problem

As it can be seen, skin friction can be calculated as:

$$C_f(s) = \frac{2\tau_w(s)}{\rho_e(s)U_e^2(s)} \quad (141)$$

Equation (134) is a definition rather than a closure: evaluating it still requires τ_w . The Quick mode estimates wall shear with an Eckert-reference-temperature turbulent flat-plate correlation and the assumed one-seventh-power profile. This deliberately weak and inexpensive closure is used only for screening. The recommended final calculation obtains τ_w from the molecular wall gradient resolved by the BLIMP-inspired model described in Section 6.

6 Adiabatic BLIMP-Inspired Skin Friction Model

6.1 Scope and modelling hierarchy

The nozzle boundary layer is evaluated with a reduced-order procedure inspired by the JANNAF BLIMP-J methodology (NASA-CR-144046) [5]. The implementation is referred to here as *BLIMP-lite*: it retains a wall-normal profile, axisymmetric continuity, the compressible thin-layer momentum equation and the Cebeci–Smith turbulent closure, but it does not reproduce BLIMP-J’s chemically reacting energy and species equations and is not a reimplement of BLIMP-J. It is intended for attached-flow preliminary design and geometry optimization at substantially lower cost than a RANS calculation.

At each wall station, the quasi-one-dimensional core-flow model supplies the edge quantities

$$U_e(s), \quad p_e(s), \quad T_e(s), \quad \rho_e(s), \quad M_e(s), \quad \gamma_e(s), \quad \mu_e(s), \quad Pr_e(s), \quad (142)$$

where s is distance along the wall and y is the inward wall-normal coordinate. RocketCEA chamber, throat and exit properties are interpolated locally. The wall contour supplies the axisymmetric radius $r(s, y)$.

6.2 Adiabatic wall condition

Wall temperature is not prescribed as an arbitrary fixed input. Without a conjugate wall-conduction and coolant model, a fixed gas-side value cannot be predicted independently of the final geometry. The present gas-side calculation therefore adopts an adiabatic wall,

$$\boxed{T_w(s) = T_r(s) = T_{aw}(s)}, \quad \boxed{q_w(s) = 0}. \quad (143)$$

For the fully turbulent closure, the local recovery factor and adiabatic-wall temperature are

$$r_f(s) = Pr_e(s)^{1/3}, \quad (144)$$

$$\boxed{T_{aw}(s) = T_e(s) \left[1 + r_f(s) \frac{\gamma_e(s) - 1}{2} M_e(s)^2 \right]}. \quad (145)$$

The wall-normal temperature profile is closed using the adiabatic Walz relation,

$$\boxed{T(s, y) = T_{aw}(s) - [T_{aw}(s) - T_e(s)] \left(\frac{u(s, y)}{U_e(s)} \right)^2}. \quad (146)$$

Equation (146) satisfies $T = T_{aw}$ and $\partial T / \partial y = 0$ at the wall, and $T = T_e$ at the boundary-layer edge. Density is evaluated locally from

$$\rho(s, y) = \frac{p_e(s)}{R(s)T(s, y)}, \quad (147)$$

while molecular viscosity is obtained by log-interpolation of the RocketCEA transport values, with a mild temperature power-law extrapolation outside their tabulated interval.

6.3 Axisymmetric compressible boundary-layer equations

The thin-layer continuity equation is

$$\boxed{\frac{\partial}{\partial s}(\rho u r) + \frac{\partial}{\partial y}(\rho v r) = 0}, \quad (148)$$

and is integrated from the impermeable wall to obtain the normal velocity v , with $v_w = 0$. The streamwise momentum equation is

$$\boxed{\rho u \frac{\partial u}{\partial s} + \rho v \frac{\partial u}{\partial y} = -\frac{dp_e}{ds} + \frac{1}{r} \frac{\partial}{\partial y} \left[r (\mu + \mu_t) \frac{\partial u}{\partial y} \right]}. \quad (149)$$

The inviscid edge solution supplies the pressure-gradient forcing, equivalently

$$-\frac{dp_e}{ds} = \rho_e U_e \frac{dU_e}{ds}. \quad (150)$$

The boundary conditions are

$$u(s, 0) = 0, \quad v(s, 0) = 0, \quad u(s, y_e) = U_e(s), \quad T(s, y_e) = T_e(s). \quad (151)$$

6.4 Cebeci–Smith turbulent closure

Reynolds shear is represented through the Boussinesq eddy-viscosity hypothesis,

$$-\rho \overline{u'v'} = \mu_t \frac{\partial u}{\partial y}. \quad (152)$$

The inner Cebeci–Smith layer uses a Van-Driest-damped mixing length [7, 6],

$$\ell = \kappa y \left[1 - \exp \left(-\frac{y^+}{A^+} \right) \right], \quad \kappa = 0.40, \quad A^+ = 26, \quad (153)$$

where

$$y^+ = \frac{\rho_w u_\tau y}{\mu_w}, \quad u_\tau = \sqrt{\frac{\tau_w}{\rho_w}}. \quad (154)$$

The corresponding inner eddy viscosity is

$$\mu_{t,i} = \rho \ell^2 \left| \frac{\partial u}{\partial y} \right|. \quad (155)$$

The outer layer is represented by the displacement-thickness wake scale

$$\mu_{t,o} = 0.0168 \rho_e U_e \delta^* F_K(y), \quad (156)$$

with the Klebanoff intermittency function

$$F_K(y) = \left[1 + 5.5 \left(\frac{y}{6\delta^*} \right)^6 \right]^{-1}. \quad (157)$$

The two regions are joined algebraically according to

$$\boxed{\mu_t = \min(\mu_{t,i}, \mu_{t,o})}. \quad (158)$$

Van Driest damping ensures $\mu_t \rightarrow 0$ at the wall. Consequently, wall shear is obtained from the resolved molecular gradient rather than from an imposed flat-plate skin-friction correlation:

$$\boxed{\tau_w = \mu_w \left. \frac{\partial u}{\partial y} \right|_w}, \quad \boxed{C_f = \frac{2\tau_w}{\rho_e U_e^2}}. \quad (159)$$

Since y^+ , μ_t , the temperature-dependent properties and τ_w are mutually dependent, they are updated iteratively at every streamwise station.

6.5 Integral thicknesses and viscous losses

The density-weighted compressible displacement and momentum thicknesses are

$$\delta^*(s) = \int_0^{y_e} \left(1 - \frac{\rho u}{\rho_e U_e} \right) dy, \quad (160)$$

$$\theta(s) = \int_0^{y_e} \frac{\rho u}{\rho_e U_e} \left(1 - \frac{u}{U_e} \right) dy, \quad (161)$$

and $H = \delta^*/\theta$. Displacement blockage is introduced through

$$r_{eff}(s) = r_w(s) - \delta^*(s), \quad A_{eff}(s) = \pi r_{eff}(s)^2. \quad (162)$$

Skin friction is retained as a distinct axial force. Since $dA_w = 2\pi r ds$ and the axial tangent projection is dx/ds ,

$$F_{fric} = \int_{x_c}^{x_e} 2\pi r_w(x) \tau_w(x) dx, \quad C_{F,fric} = \frac{F_{fric}}{p_c A_t}. \quad (163)$$

The performance closure therefore becomes

$$C_{F,eff} = \eta_{div} \eta_{BL} C_{F,mom}^{CEA} + \varepsilon \frac{p_e - p_a}{p_c} - C_{F,fric}. \quad (164)$$

Blockage and wall shear are thus represented separately and are not hidden inside a single empirical efficiency.

6.6 Numerical procedure and limitations

The equations are marched implicitly over 61 streamwise stations. Each station uses 41 wall-normal nodes clustered toward the wall. The continuity equation provides $v(y)$; the momentum equation is discretized in conservative radial form; and Picard iterations update u , T , ρ , μ , y^+ , μ_t and τ_w . The converged integral quantities are interpolated onto the complete geometry grid. In a default-case grid comparison against 91 streamwise by 57 wall-normal nodes, exit C_f and integrated $C_{F,fric}$ changed by less than 0.5%, while exit δ^* changed by approximately 4%.

The model assumes an attached, fully turbulent, axisymmetric boundary layer with no wall mass transfer. The inlet profile is an engineering initialization because the upstream chamber boundary-layer history is not otherwise available. Transition, relaminarization, shock–boundary-layer interaction and separated flow are not predicted. The adiabatic Walz relation [8] replaces the full BLIMP-J reacting energy and species system. Cebeci–Smith is an algebraic equilibrium closure; final designs therefore still require mesh-refined RANS/CFD and experimental validation.

7 Loss-Corrected Quasi-One-Dimensional Model for Geometry Optimization

7.1 Reduced-order performance closure

7.1.1 Need for a loss-corrected objective

The ideal quasi-one-dimensional model developed in Section 3 provides the core-flow state, but it does not account for the loss of useful axial momentum caused by flow divergence and viscous blockage. These effects are included through the momentum efficiency

$$\eta_{mom} = \eta_{div}\eta_{BL}, \quad (165)$$

where η_{div} is the divergence efficiency and η_{BL} is the boundary-layer velocity efficiency.

7.1.2 Exit-flow divergence correction

The divergence efficiency is evaluated from the exit-wall angle supplied by the geometric model of Section 8. The axisymmetric approximation used by the simulator is

$$\eta_{div} = \frac{1 + \cos \theta_{out}}{2}. \quad (166)$$

This factor approaches unity as the exit flow becomes parallel to the nozzle axis and decreases as θ_{out} increases. The value of θ_{out} is a derived result of the contour formulation and is not an independent optimization variable.

7.1.3 Boundary-layer correction

The boundary-layer quantities are calculated with either the Quick screening model of Section 5 or the BLIMP-inspired model of Section 6; their equations are not repeated here. The blockage contribution enters through

$$\eta_{BL} = \frac{U_{e,eff}}{U_{e,ist}}, \quad 0 \leq \eta_{BL} \leq 1, \quad (167)$$

where $U_{e,eff}$ is the corrected exit velocity and $U_{e,ist}$ is the uncorrected quasi-one-dimensional value. Thus, $\eta_{BL} = 1$ represents no predicted viscous velocity loss, whereas lower values represent increasing boundary-layer blockage. The wall-shear contribution remains a separate force coefficient, $C_{F,fric}$, defined in Eq. (163); it is not hidden inside η_{BL} .

7.1.4 Effective ambient thrust coefficient

RocketCEA supplies the ideal momentum thrust coefficient $C_{F,mom}^{CEA}$ for each operating point and expansion ratio. The corrected momentum contribution is

$$C_{F,mom}^{eff} = \eta_{div}\eta_{BL}C_{F,mom}^{CEA}. \quad (168)$$

The ambient pressure contribution is evaluated separately:

$$C_{F,p} = \varepsilon \frac{p_e - p_a}{p_c}. \quad (169)$$

The pressure term is positive for underexpanded operation, zero when $p_e = p_a$, and negative for overexpanded attached operation. It is not multiplied by the loss efficiencies, since $\eta_{div}\eta_{BL}$ is applied only to the momentum term. The effective ambient thrust coefficient is therefore

$$C_{F,eff} = \eta_{div}\eta_{BL}C_{F,mom}^{CEA} + \varepsilon \frac{p_e - p_a}{p_c} - C_{F,fric}. \quad (170)$$

The corresponding thrust estimate is

$$F_{eff} = C_{F,eff}p_cA_t. \quad (171)$$

Since chamber pressure and throat area remain fixed during an optimization run, maximizing $C_{F,eff}$ is equivalent to maximizing F_{eff} . The objective therefore accounts simultaneously for divergence loss, boundary-layer blockage, integrated wall friction and ambient pressure mismatch. The dependence of p_e on ε , including the ideal condition $p_e = p_a$, follows from the RocketCEA exit solution and Eq. (5).

Ambient pressure is entered in the simulator in atmospheres and converted internally according to

$$p_a[\text{bar}] = 1.01325 p_a[\text{atm}]. \quad (172)$$

All pressures used in Eq. (170) are internally converted to consistent units.

7.1.5 Separated-flow handling and scope

Equation (169) assumes attached flow to the exit plane. If RocketCEA classifies a candidate as separated, the candidate is assigned zero fitness because the adopted exit-plane pressure term is no longer applicable. The resulting loss correction remains a reduced-order engineering closure and must be validated with higher-fidelity analysis or experimental data for final design use.

8 Rao/Sutton-Inspired Parametric Bell-Nozzle Geometry Formulation

8.1 Analytical contour definition

8.1.1 Independent geometric design parameters

The nozzle geometry is defined by a reduced set of parameters:

- throat radius, r_t ;
- chamber radius, r_c ;
- expansion ratio, ε ;
- subsonic converging angle, θ_{sub} ;
- initial supersonic angle, θ_{in} ;
- reference conical half-angle, α ;
- bell length fraction, K_b , defined as a percentage of the reference conical length.

The geometric model follows classical bell-nozzle proportions described by Sutton and Biblarz, with circular throat arcs and a quadratic diverging section. It is a Rao/Sutton-inspired parametric contour, not a thrust-optimized Rao TOP contour obtained from a method-of-characteristics solution. The contour is axisymmetric, so the two-dimensional function $r(x)$ fully defines the three-dimensional surface by revolution around the nozzle axis.

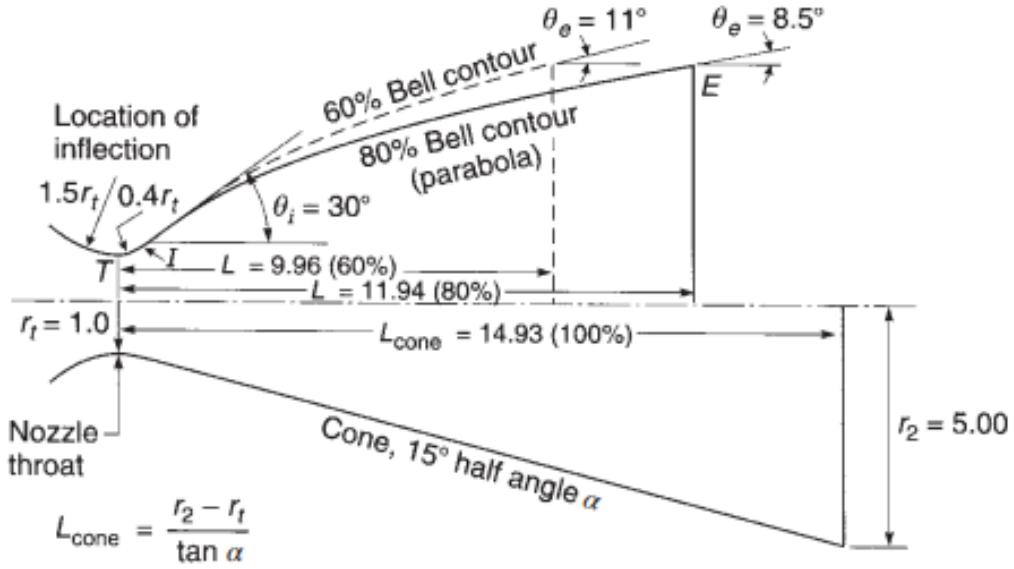


Figure 6: Reference bell-nozzle proportions from the Sutton/Biblarz formulation.

8.1.2 Global piecewise contour architecture

The contour is constructed from four segments:

1. initial straight converging line;
2. subsonic circular arc upstream of the throat;
3. supersonic circular arc immediately downstream of the throat;
4. parabolic supersonic contour up to the exit plane.

The coordinate origin is initially placed at the throat, with the nozzle axis along x and the local radius along r . The throat point is therefore

$$(x_t, r_t) = (0, r_t). \quad (173)$$

The full profile is later translated horizontally so that the exported and displayed geometry starts at $x = 0$.

8.1.3 Upstream throat-radius circular arc

The subsonic circular arc uses a radius

$$R_{sub} = 1.5r_t. \quad (174)$$

Its center is placed on the radial axis at

$$C_{sub} = (0, 2.5r_t). \quad (175)$$

The circular equation is

$$x^2 + (r - 2.5r_t)^2 = (1.5r_t)^2. \quad (176)$$

The lower branch of the circle defines the desired arc:

$$r(x) = 2.5r_t - \sqrt{(1.5r_t)^2 - x^2}, \quad x \in [x_r, 0]. \quad (177)$$

The connection point between the initial straight line and the subsonic arc is

$$x_r = -1.5r_t \sin(\theta_{sub}), \quad (178)$$

$$r_r = 2.5r_t - 1.5r_t \cos(\theta_{sub}). \quad (179)$$

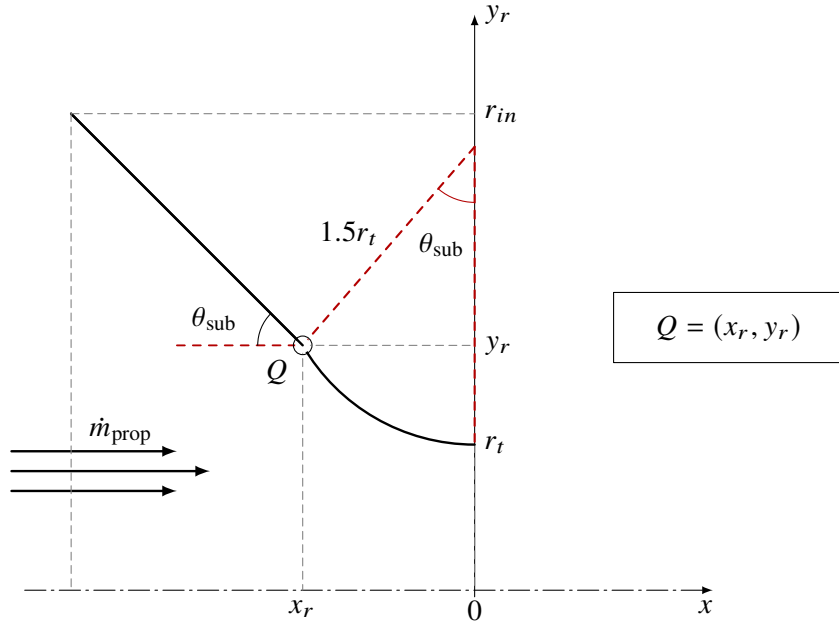


Figure 7: Subsonic geometric relations and definition of the converging arc.

8.1.4 Linear converging-section approximation

The initial converging section is approximated as a straight line tangent to the subsonic arc at (x_r, r_r) . Its slope is set by the angle θ_{sub} :

$$m_{sub} = -\tan(\theta_{sub}). \quad (180)$$

The line equation is

$$r(x) = m_{sub}x + b_{sub}, \quad (181)$$

where

$$b_{sub} = r_r - m_{sub}x_r. \quad (182)$$

Substituting Eqs. (178) and (179), the line can also be written as

$$r(x) = -\tan(\theta_{sub})x + r_t [2.5 - 1.5 \cos(\theta_{sub}) - 1.5 \sin(\theta_{sub}) \tan(\theta_{sub})]. \quad (183)$$

The first axial coordinate x_f is obtained from the chamber radius condition,

$$r(x_f) = r_c, \quad (184)$$

which gives

$$x_f = \frac{r_c - b_{sub}}{m_{sub}}. \quad (185)$$

The contraction ratio is then

$$CR = \frac{A_c}{A_t} = \frac{r_c^2}{r_t^2}. \quad (186)$$

8.1.5 Downstream throat-radius circular arc

Immediately downstream of the throat, the supersonic arc uses radius

$$R_{sup} = 0.4r_t. \quad (187)$$

Its center is placed at

$$C_{sup} = (0, 1.4r_t). \quad (188)$$

The circular equation is

$$x^2 + (r - 1.4r_t)^2 = (0.4r_t)^2. \quad (189)$$

The branch used for the nozzle profile is

$$r(x) = 1.4r_t - \sqrt{(0.4r_t)^2 - x^2}, \quad x \in [0, x_P]. \quad (190)$$

Equivalently, the arc can be parameterized by the angle θ :

$$x(\theta) = 0.4r_t \cos \theta, \quad (191)$$

$$r(\theta) = 1.4r_t - 0.4r_t \sin \theta, \quad (192)$$

with

$$\theta \in \left[\frac{\pi}{2}, \frac{\pi}{2} - \theta_{in} \right]. \quad (193)$$

The final point of the arc is the beginning of the parabolic contour:

$$x_P = 0.4r_t \sin(\theta_{in}), \quad (194)$$

$$r_P = 1.4r_t - 0.4r_t \cos(\theta_{in}). \quad (195)$$

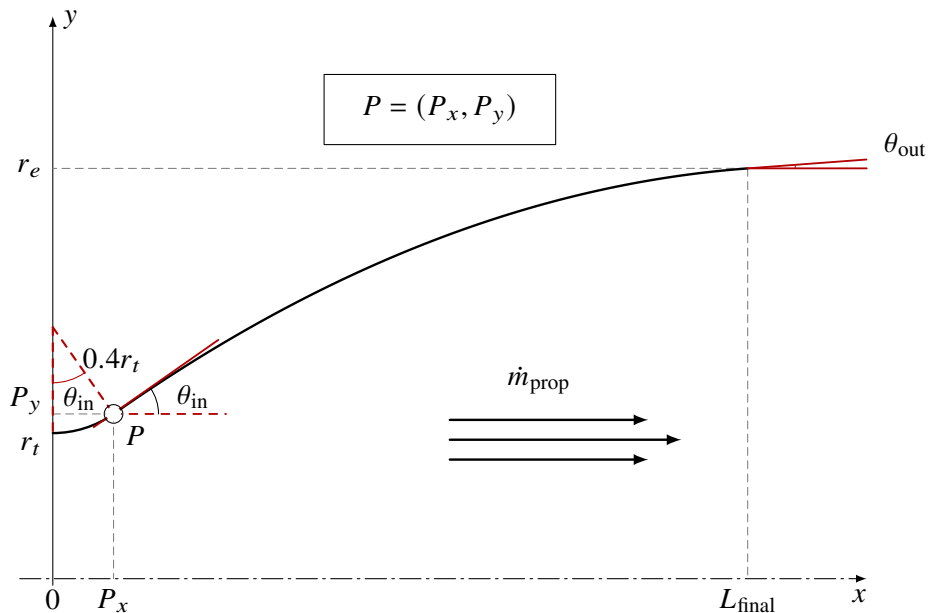


Figure 8: Supersonic throat arc and transition into the parabolic diverging contour.

8.1.6 Reference conical length and bell shortening factor

The reference conical nozzle length is computed from the exit radius, throat radius and conical half-angle α :

$$L_{cone} = \frac{r_e - r_t}{\tan \alpha}. \quad (196)$$

The bell nozzle uses only a fraction K_b of this length:

$$L_{parab} = K_b L_{cone}. \quad (197)$$

The final axial coordinate of the parabolic section in the throat-centered coordinate system is

$$L = x_P + L_{parab}. \quad (198)$$

8.1.7 Parabolic diverging-contour closure

The parabolic contour is written as

$$r(x) = ax^2 + bx + c, \quad x \in [x_P, L]. \quad (199)$$

A general parabola has three coefficients, so three independent conditions are imposed:

$$r(x_P) = r_P, \quad (200)$$

$$r'(x_P) = \tan(\theta_{in}), \quad (201)$$

$$r(L) = r_e. \quad (202)$$

The first condition enforces positional continuity with the supersonic arc. The second condition enforces slope continuity and guarantees a smooth transition between the circular arc and the parabola. The third condition fixes the exit radius and therefore the desired expansion ratio.

The ideal case could also impose an exit angle condition,

$$r'(L) = \tan(\theta_{out}), \quad (203)$$

but this would overdefine a simple parabola. Therefore, in the present formulation, θ_{out} is not prescribed; it is an output of the geometry.

The coefficient system is

$$\begin{bmatrix} x_P^2 & x_P & 1 \\ 2x_P & 1 & 0 \\ L^2 & L & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} r_P \\ \tan(\theta_{in}) \\ r_e \end{bmatrix}. \quad (204)$$

Solving this linear system gives the parabolic coefficients that fully define the diverging section.

8.1.8 Exit-plane wall angle

The exit angle is obtained from the slope of the parabolic contour evaluated at the nozzle exit plane:

$$\theta_{out} = \arctan(2aL + b). \quad (205)$$

This angle represents the local divergence of the flow at the nozzle exit and directly affects axial momentum efficiency. A large exit angle increases the radial velocity component and reduces useful axial thrust.

8.1.9 Complete piecewise radius function

Using the previous relations, the complete throat-centered nozzle contour can be written as

$$r(x) = \begin{cases} -\tan(\theta_{sub})x + b_{sub}, & x \in [x_f, x_r], \\ 2.5r_t - \sqrt{(1.5r_t)^2 - x^2}, & x \in [x_r, 0], \\ 1.4r_t - \sqrt{(0.4r_t)^2 - x^2}, & x \in [0, x_P], \\ ax^2 + bx + c, & x \in [x_P, L]. \end{cases} \quad (206)$$

This equation is one of the most important outputs of the geometric formulation because it gives a continuous mathematical representation of the entire nozzle contour.

8.1.10 Geometric station and angle summary

An illustrative nozzle geometry corresponding to $\varepsilon = 5.0$ and $r_t = 0.020$ m is presented in Fig. 9 to identify the stations and angles introduced in the previous sections.

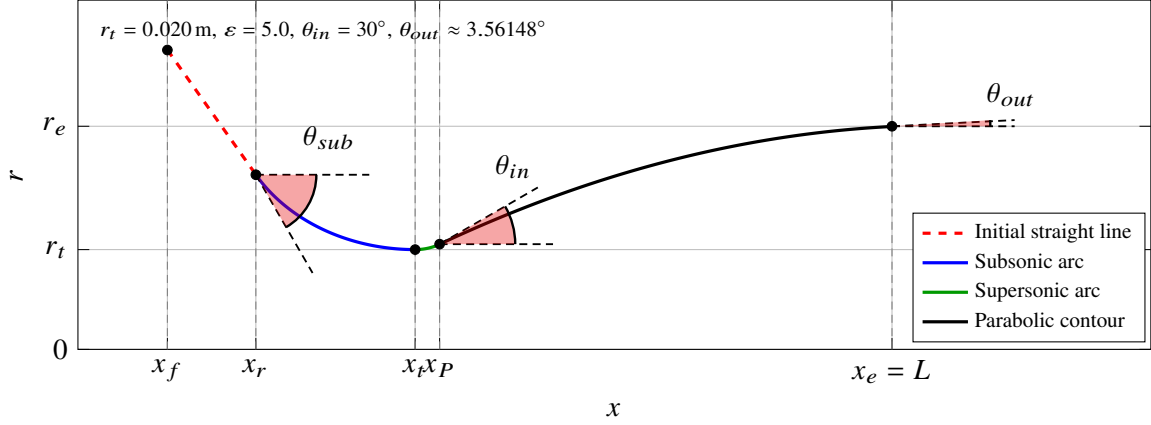


Figure 9: Two-dimensional nozzle contour with region definitions, stations and angle notation.

8.2 Numerical Coordinate Conditioning and Two-Dimensional Contour Output

8.2.1 Axial coordinate translation

The throat-centered coordinate system is convenient for mathematical construction, but it produces negative x values in the subsonic converging section. For plotting, CAD export and easier interpretation, the profile is translated horizontally so that the first point of the nozzle starts at $x = 0$.

The required translation is

$$\Delta x = -x_f. \quad (207)$$

Using the line geometry, this can also be written as

$$\Delta x = -x_r + \frac{r_c - y_r}{\tan(\theta_{sub})}. \quad (208)$$

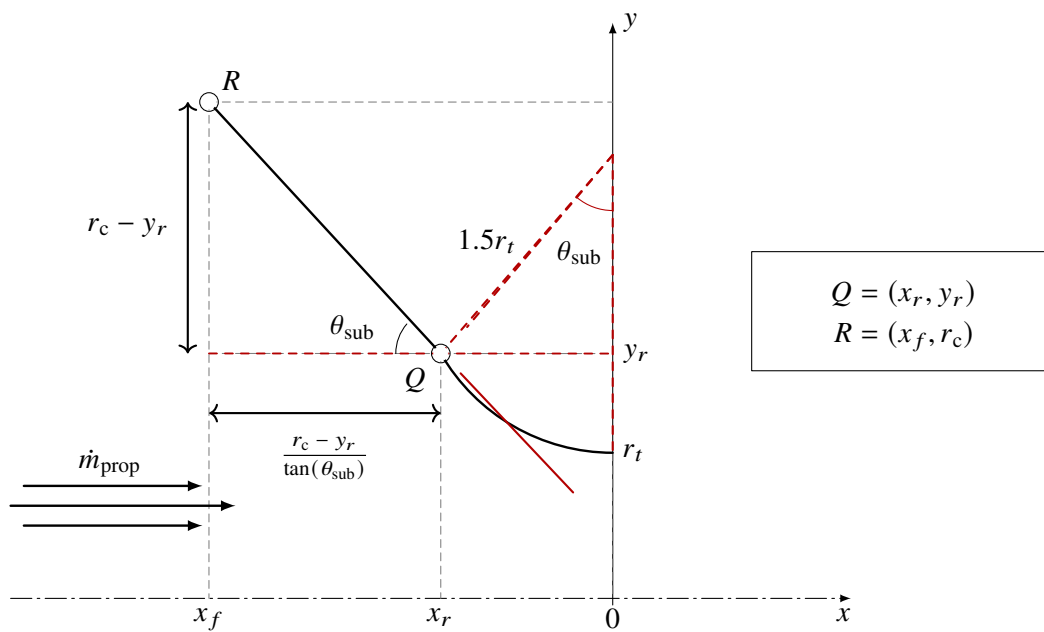


Figure 10: Horizontal translation used to move the nozzle start to $x = 0$.

After translation, all axial coordinates become

$$\bar{x} = x + \Delta x. \quad (209)$$

The translated domain is

$$\bar{x} \in [0, L - x_f]. \quad (210)$$

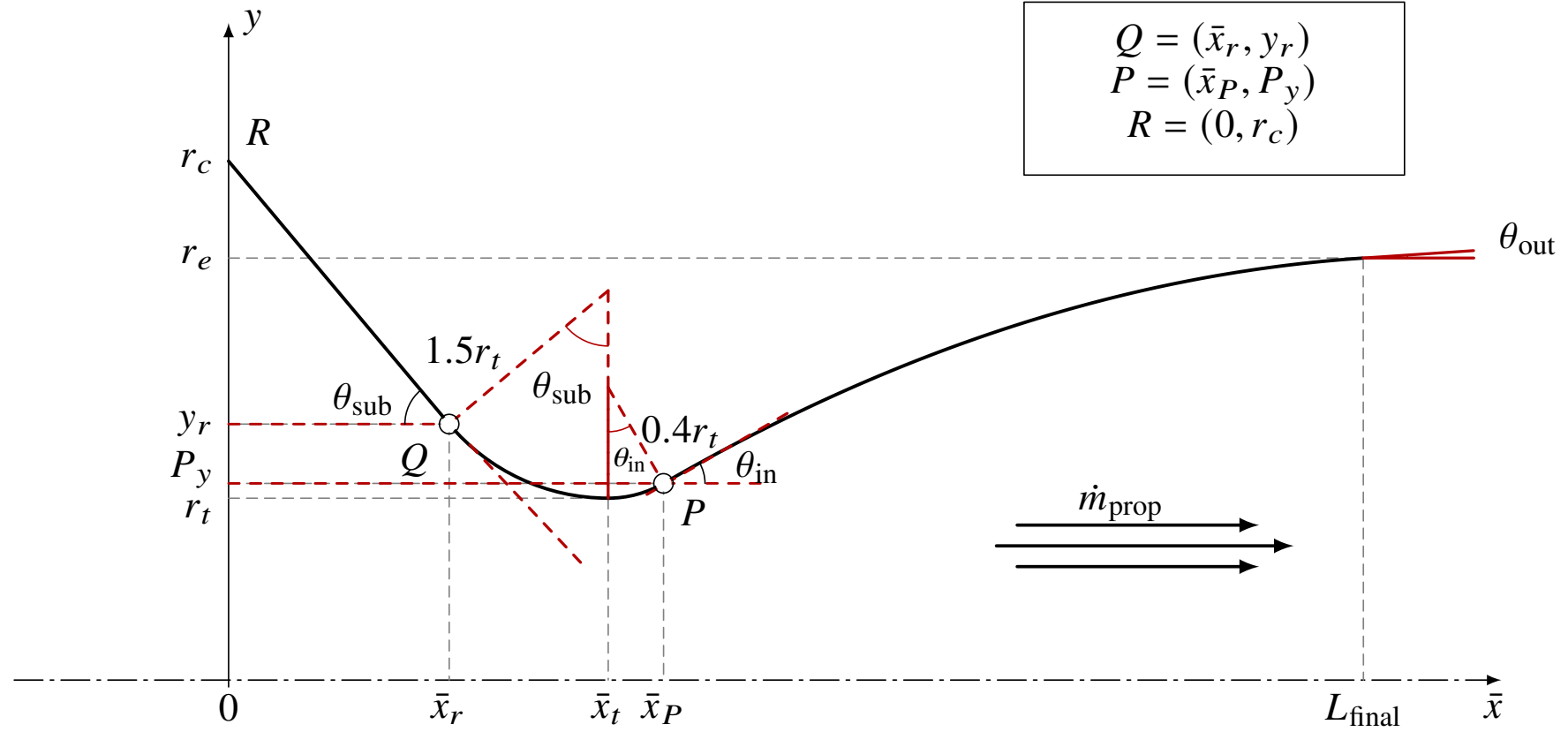


Figure 11: Combined translated nozzle contour showing the straight converging section, the subsonic throat arc, the supersonic throat arc, and the parabolic diverging section in the translated domain.

8.2.2 Derived geometric outputs

The simulator outputs several secondary quantities that help evaluate the geometry:

$$CR = \frac{r_c^2}{r_t^2}, \quad (211)$$

$$\varepsilon = \frac{r_e^2}{r_t^2}, \quad (212)$$

$$L_{nozzle} = L - x_f, \quad (213)$$

$$\theta_{out} = \arctan(2aL + b). \quad (214)$$

These quantities are displayed together with the two-dimensional profile to make each generated geometry immediately interpretable.

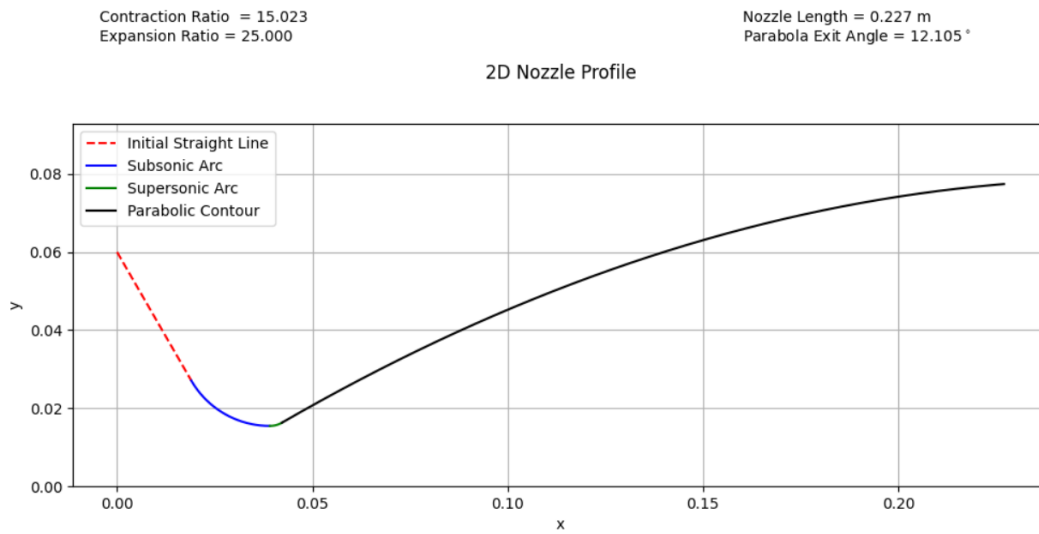


Figure 12: Example 1 of a generated two-dimensional nozzle geometry profile.

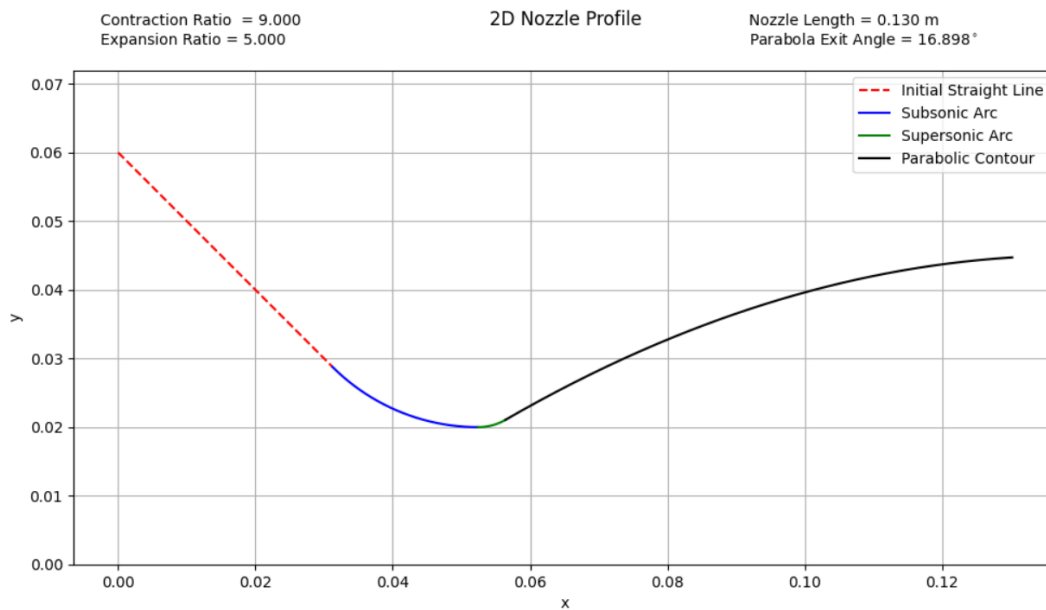


Figure 13: Example 2 of a generated two-dimensional nozzle geometry profile.

8.3 Axisymmetric Three-Dimensional Geometry Reconstruction

8.3.1 Surface-of-revolution definition

Because the nozzle is axisymmetric, the three-dimensional surface is generated by revolving the two-dimensional radius function around the nozzle axis. For each axial coordinate x_i and radius r_i , a circular ring is generated using the angular coordinate ϕ :

$$X_i(\phi) = x_i, \quad (215)$$

$$Y_i(\phi) = r_i \cos \phi, \quad (216)$$

$$Z_i(\phi) = r_i \sin \phi, \quad (217)$$

with

$$\phi \in [0, 2\pi]. \quad (218)$$

The complete nozzle surface is therefore

$$\mathbf{S}(x, \phi) = \begin{bmatrix} x \\ r(x) \cos \phi \\ r(x) \sin \phi \end{bmatrix}. \quad (219)$$

This formulation is important because it makes clear that the three-dimensional geometry is not an independent model. It is the direct revolution of the piecewise two-dimensional contour from Eq. (206).

8.3.2 Structured mesh matrices for visualization

For numerical visualization, the axial and angular coordinates are assembled into matrices. If

$$\mathbf{x} = [x_0 \quad x_1 \quad \cdots \quad x_n]^T, \quad (220)$$

and

$$\boldsymbol{\phi} = [\phi_0 \quad \phi_1 \quad \cdots \quad \phi_m], \quad (221)$$

then the axial mesh has the structure

$$\mathbf{X} = \begin{bmatrix} x_0 & x_0 & \cdots & x_0 \\ x_1 & x_1 & \cdots & x_1 \\ \vdots & \vdots & \ddots & \vdots \\ x_n & x_n & \cdots & x_n \end{bmatrix}. \quad (222)$$

The angular mesh is

$$\boldsymbol{\Phi} = \begin{bmatrix} \phi_0 & \phi_1 & \cdots & \phi_m \\ \phi_0 & \phi_1 & \cdots & \phi_m \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0 & \phi_1 & \cdots & \phi_m \end{bmatrix}. \quad (223)$$

The radial mesh is

$$\mathbf{R} = \begin{bmatrix} r_0 & r_0 & \cdots & r_0 \\ r_1 & r_1 & \cdots & r_1 \\ \vdots & \vdots & \ddots & \vdots \\ r_n & r_n & \cdots & r_n \end{bmatrix}. \quad (224)$$

The Cartesian surface matrices are finally

$$Y = R \cos \Phi, \quad (225)$$

$$Z = R \sin \Phi. \quad (226)$$

8.3.3 Three-dimensional geometry inspection

The three-dimensional plot is used as a quick visual validation before exporting the profile to CAD. It helps identify discontinuities, wrong radius values, exaggerated exit radius, incorrect translations, etc.

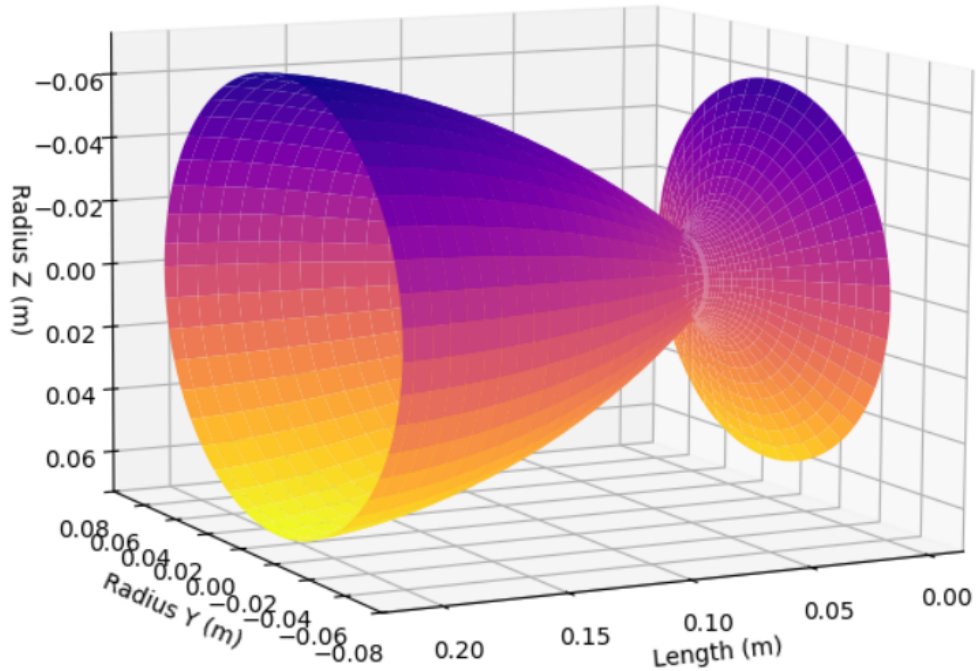


Figure 14: Example of three-dimensional nozzle visualization generated by surface revolution.

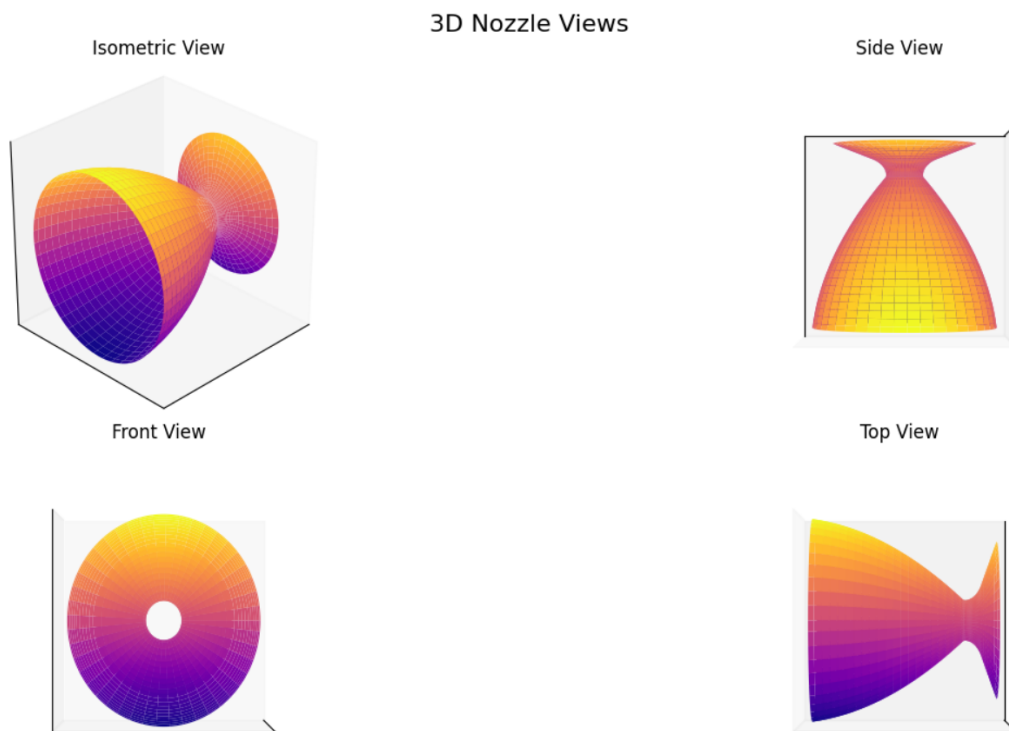


Figure 15: Example of multi-view visualization of the generated nozzle geometry.

8.4 CAD-Oriented Geometry Export and Manufacturing Workflow

8.4.1 CAD-compatible coordinate export

One of the practical objectives of the simulator is to convert the numerically generated nozzle contour into a format suitable for computer-aided design (CAD). The exported coordinates describe the two-dimensional meridional profile of the nozzle and can be imported into software such as Autodesk Fusion or SolidWorks.

The exported point set is written as

$$\mathcal{P} = \{(x_i, r_i, 0)\}_{i=0}^N, \quad (227)$$

where x_i is the axial coordinate and r_i is the corresponding wall radius. The third coordinate is set to zero because the contour is imported as a planar curve. Before export, the axial coordinates are translated so that the first point of the contour is located at $x = 0$, and the coordinates are converted from metres to millimetres.

The resulting curve represents the internal wall of the nozzle. A three-dimensional body can subsequently be reconstructed in CAD by defining the required wall thickness, closing the sketch and revolving the resulting cross-section around the nozzle axis.

8.4.2 Point reduction and sketch-conditioning strategy

The numerical geometry uses a relatively dense point distribution to provide smooth plots and sufficiently resolved flow, thermal and boundary-layer profiles. Directly importing this complete point set into CAD, however, may produce an unnecessarily complex spline. Excessive point density, particularly near segment junctions, can lead to local oscillations, poor spline conditioning and difficulties when applying sketch constraints.

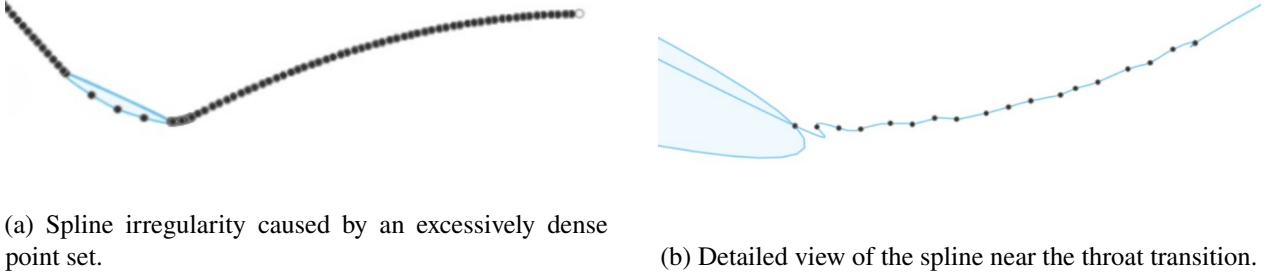


Figure 16: CAD sketch irregularities associated with an unsuitable point distribution.

The initial CAD-export implementation therefore included a two-stage point-conditioning procedure. First, a target of approximately 100 points was distributed proportionally among the four geometric regions:

1. the straight convergent section;
2. the subsonic throat arc;
3. the supersonic throat arc;
4. the parabolic bell contour.

A minimum of two points was retained in each region so that no geometric segment was omitted. Within each segment, points were selected at approximately uniform index intervals.

A second sequential filter was then applied to remove clustered points. A candidate point was retained only when its distance from the previously accepted point satisfied

$$d_i = \sqrt{(x_i - x_k)^2 + (r_i - r_k)^2} \geq d_{\min}, \quad (228)$$

where k denotes the last accepted point and $d_{\min}$ is the prescribed minimum spacing. A value of approximately $d_{\min} = 1$ mm was used during the CAD-conditioning tests. The first and final contour points, together with the points defining the main geometric transitions, should be preserved explicitly to maintain the chamber inlet, throat and nozzle-exit dimensions.

The current modular implementation exports the complete set of unique geometry points to `geometry.csv`. The reduction procedure described above corresponds to the CAD-conditioning stage developed in the initial version and can be applied as an additional preprocessing step when a lighter CAD spline is required.

8.4.3 CAD reconstruction workflow

The complete geometry-reconstruction workflow is summarized as follows:

1. generate the two-dimensional contour using the geometric model;
2. translate the axial coordinates so that the contour begins at $x = 0$;
3. remove duplicate points at the junctions between geometric segments;
4. reduce and filter the point set when CAD conditioning is required;
5. convert the coordinates from metres to millimetres;
6. export the coordinates as a three-column point file;
7. import the point file into the CAD environment;
8. create a spline through the imported points;
9. define the wall thickness by offsetting the internal contour;
10. close the cross-section and revolve it around the nozzle axis.

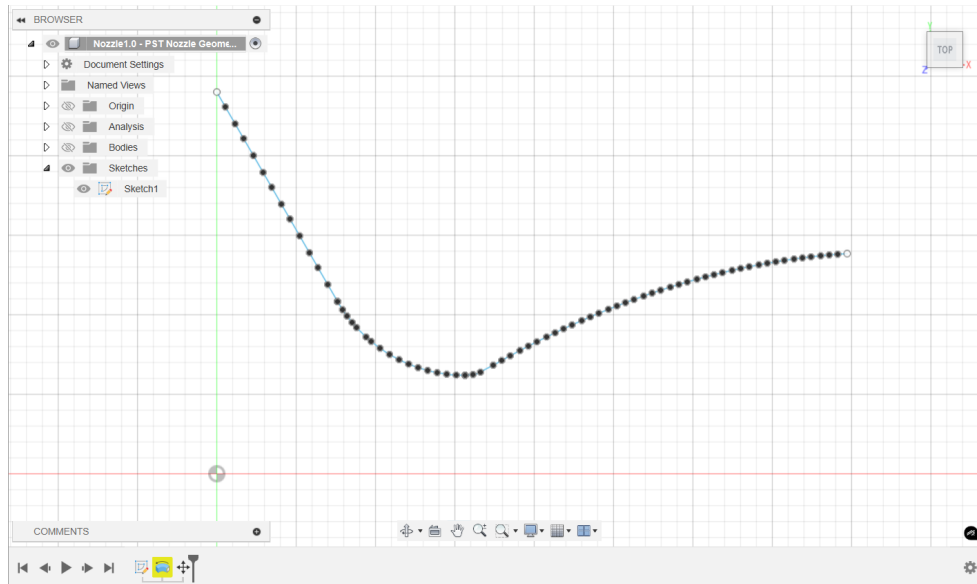
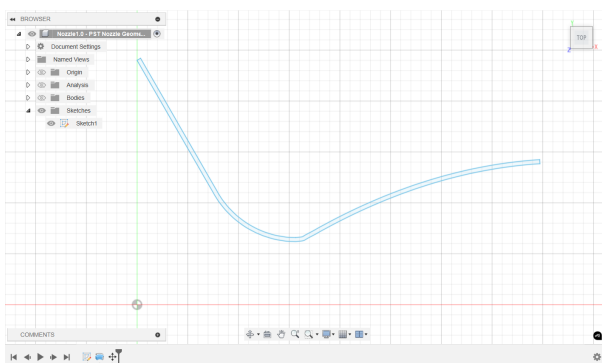
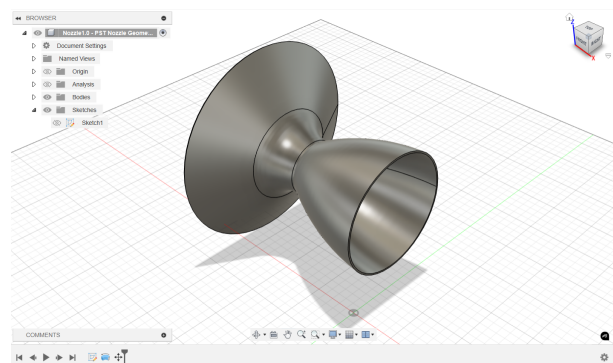


Figure 17: Nozzle contour reconstructed in CAD using the conditioned point distribution.



(a) Definition of the nozzle wall thickness.



(b) Final axisymmetric nozzle body after revolution.

Figure 18: CAD operations used to convert the exported meridional contour into a manufacturable three-dimensional body.

9 Genetic-Algorithm Optimization Framework

9.1 Optimization problem definition

The genetic algorithm searches the prescribed design space while retaining the engine operating point and pre-sizing quantities supplied by the user. The detailed quasi-one-dimensional, boundary-layer, performance and contour formulations are given in Sections 3–8 and are not repeated here. Within the optimizer, these models are treated as a coupled candidate-evaluation function

$$S : (q, x_{fixed}) \longrightarrow C_{F,eff}, \quad (229)$$

where q is the chromosome and x_{fixed} contains the quantities held constant during one optimization run.

For every candidate, the simulator evaluates the required RocketCEA properties, the common physical models and the effective thrust coefficient defined in Eq. (170). The genetic algorithm therefore operates on the same physical model used by the simulator rather than on an independent surrogate.

The Bartz heat-transfer profile is omitted from the repeated fitness calculation because thermal loading is not part of the current objective. No wall-temperature input is passed to the evaluator: both boundary-layer closures use the locally derived adiabatic-wall temperature of Eq. (37).

Two evaluation modes are exposed without changing the chromosome or GA operators. *Quick screening* uses the weak integral closure of Section 5; *BLIMP-lite* uses the profile marcher of Section 6. A solution obtained through Quick screening is regenerated and reported with BLIMP-lite before it is accepted as the selected geometry.

9.2 Chromosome and fixed quantities

9.2.1 Chromosome definition

The implemented chromosome contains three real-valued genes:

$$q = [\varepsilon \quad K_b \quad \theta_{in}]^T, \quad (230)$$

where ε is the exit-to-throat area ratio, K_b is the bell-length fraction and θ_{in} is the initial divergent-wall angle. Their default search bounds are

$$2.0 \leq \varepsilon \leq 15.0, \quad (231)$$

$$0.60 \leq K_b \leq 1.00, \quad (232)$$

$$20^\circ \leq \theta_{in} \leq 35^\circ. \quad (233)$$

All genes are represented as floating-point values. The limits can be changed directly from the optimization interface.

The exit angle θ_{out} is not a gene because it is a derived output of the contour model. The convergent angle θ_{sub} is also excluded from the chromosome and remains fixed during the search.

9.2.2 Fixed quantities

The fixed input vector may be represented as

$$x_{fixed} = [p_c \quad O/F \quad p_a \quad r_t \quad D_c \quad \alpha \quad \theta_{sub}]^T. \quad (234)$$

The default values used by the interface are listed in Table 1.

The recovery factor is not included because it is derived locally from the interpolated RocketCEA Prandtl number rather than supplied as an independent input. Since ε is part of the chromosome, the exit properties returned by RocketCEA are candidate dependent and cannot be treated as a single common result for the complete population.

Table 1: Default inputs and their roles during optimization.

Quantity	Symbol	Default value	GA status
Chamber pressure	p_c	30 bar	Fixed
Mixture ratio	O/F	6.5	Fixed
Ambient pressure	p_a	1.0 atm	Fixed
Expansion ratio	ε	5.6	Optimized
Throat radius	r_t	0.01728 m	Fixed
Chamber diameter	D_c	0.120 m	Fixed
Reference cone half-angle	α	15°	Fixed
Bell-length fraction	K_b	0.80	Optimized
Initial divergent angle	θ_{in}	30°	Optimized
Convergent angle	θ_{sub}	50°	Fixed

9.3 Fitness-function definition

The scalar objective maximized by the genetic algorithm is the effective ambient thrust coefficient introduced in Section 7:

$$J(\mathbf{q}) = C_{F,eff}(\mathbf{q}). \quad (235)$$

Explicitly,

$$J(\mathbf{q}) = \eta_{div}(\mathbf{q})\eta_{BL}(\mathbf{q})C_{F,mom}^{CEA}(\mathbf{q}) + \varepsilon(\mathbf{q})\frac{p_e(\mathbf{q}) - p_a}{p_c} - C_{F,fric}(\mathbf{q}). \quad (236)$$

The objective is maximized directly, so no sign inversion is required. It contains a loss-corrected momentum contribution, the signed ambient pressure-thrust contribution and integrated wall friction. It does not include additional empirical, mass, manufacturing-cost or thermal-limit penalties.

9.4 Candidate validation and constraint handling

The gene bounds in Eqs. (231)–(233) provide the first level of constraint enforcement. Each generated candidate is then passed to the common simulator validation and to the models defined in Sections 3–8. The geometric admissibility conditions are part of the contour model of Section 8 and are not duplicated here.

A candidate is rejected if the common simulator reports an invalid input, an invalid contour, a singular numerical system, a non-finite result or a non-positive effective thrust coefficient. Candidates classified as separated by RocketCEA are also rejected.

Constraint handling uses a death-penalty strategy:

$$J_{GA}(\mathbf{q}) = \begin{cases} C_{F,eff}(\mathbf{q}), & \text{valid, finite and attached-flow candidate,} \\ 0, & \text{otherwise.} \end{cases} \quad (237)$$

There is no continuous penalty proportional to the magnitude of a violation.

9.5 Genetic-algorithm configuration

The optimizer is implemented using PyGAD [10]. Its default configuration is summarized in Table 2. The settings are user-configurable. The implemented validation requires

$$N_p \geq 4, \quad G_{max} \geq 1, \quad 2 \leq N_{parents} \leq N_p. \quad (238)$$

Table 2: Default configuration of the implemented genetic algorithm.

Parameter	Implemented value
Chromosome representation	Three real-valued genes
Population size	$N_p = 100$ individuals
Maximum generations	$G_{max} = 300$
Mating parents	20
Parent selection	Tournament selection
Crossover	Uniform crossover
Crossover probability	$P_c = 0.85$
Mutation	Adaptive mutation
Adaptive mutation percentages	[67%, 34%]
Elitism	Best three individuals
Saturation limit	40 generations
Default evaluation mode	Four worker processes
Repeated-candidate cache	Enabled
Saved full solution history	No
Explicit random seed	No

9.5.1 Tournament selection

Tournament selection gives preference to candidates with higher $C_{F,eff}$ while preserving stochastic exploration. Candidate parents are sampled from the current population and the best member of each tournament is selected for reproduction. With the default settings, 20 parents participate in each generation.

9.5.2 Uniform crossover

Uniform crossover constructs each offspring by selecting each gene from either parent. For example, given

$$\mathbf{q}^{(1)} = \begin{bmatrix} \varepsilon^{(1)} & K_b^{(1)} & \theta_{in}^{(1)} \end{bmatrix}^T, \quad (239)$$

$$\mathbf{q}^{(2)} = \begin{bmatrix} \varepsilon^{(2)} & K_b^{(2)} & \theta_{in}^{(2)} \end{bmatrix}^T, \quad (240)$$

one possible offspring is

$$\mathbf{q}^{(o)} = \begin{bmatrix} \varepsilon^{(1)} & K_b^{(2)} & \theta_{in}^{(1)} \end{bmatrix}^T. \quad (241)$$

Crossover is applied with probability $P_c = 0.85$.

9.5.3 Adaptive mutation

Adaptive mutation changes the mutation intensity according to candidate fitness. The configured percentages are

$$\mathbf{p}_m = \begin{bmatrix} 67\% & 34\% \end{bmatrix}. \quad (242)$$

Lower-fitness candidates receive the larger mutation percentage, promoting exploration, while higher-fitness candidates receive the smaller percentage, promoting local refinement. For the three-gene chromosome, these percentages correspond approximately to mutating two genes and one gene, respectively. All mutations remain bounded by the defined gene spaces.

9.5.4 Elitism

The best three candidates are transferred to the following generation:

$$N_{elite} = 3. \quad (243)$$

Elitism prevents the best solution found so far from being lost during crossover or mutation.

9.6 Evaluation acceleration

The optimizer supports serial, threaded and multiprocessing evaluation. The default mode uses a persistent pool of four worker processes. Each process receives an isolated RocketCEA working directory, preventing simultaneous Fortran evaluations from using the same temporary files.

An exact-match fitness cache stores previously evaluated chromosomes. When an identical chromosome reappears, its stored fitness is reused. The ideal expansion ratio associated with the fixed values of p_c , O/F and p_a is cached separately because it does not depend on the candidate chromosome.

These measures reduce evaluation time without changing population generation, selection, crossover, mutation, elitism or stopping logic.

9.7 Stopping, cancellation and reported progress

The optimization terminates when the first of the following conditions is met:

1. the maximum number of generations is reached;
2. the best fitness does not improve for the selected saturation interval, equal to 40 generations by default;
3. the user requests cancellation through the interface.

Cancellation is checked at the end of each generation. The current generation is therefore allowed to finish before execution stops.

During execution, the interface displays the completed generation, percentage, elapsed time, estimated remaining time and generation rate. It also reports the best $C_{F,eff}$ and the current chromosome. If no candidate with positive fitness is found, the optimization is declared unsuccessful.

Otherwise, the returned result is

$$\mathbf{y}_{GA} = \left[\varepsilon^* \quad K_b^* \quad \theta_{in}^* \quad \theta_{out}^* \quad p_e^* \quad C_{F,eff}^* \quad N_g \right]^T, \quad (244)$$

where N_g is the number of completed generations. The ambient operating-mode classification is also retained. Because no explicit random seed is assigned, repeated runs with identical inputs may return slightly different chromosomes.

9.8 Optimization workflow

Figure 19 summarizes the implemented algorithm without repeating the internal formulations of the physical simulator.

9.9 Algorithmic limitations and possible extensions

The implemented GA solves a single-objective, single-operating-point problem. It is stochastic and does not guarantee the global optimum. Its result also depends on the accuracy and validity range of the coupled simulator used to evaluate each chromosome.

Future algorithmic extensions may include multi-objective optimization, robustness across several ambient pressures, explicit uncertainty propagation, repeated runs with controlled random seeds, convergence-statistics reporting and surrogate-assisted evaluation. Physical or geometric extensions should be introduced in their respective model sections and then exposed to the GA only through the chromosome, constraints or candidate-evaluation function.

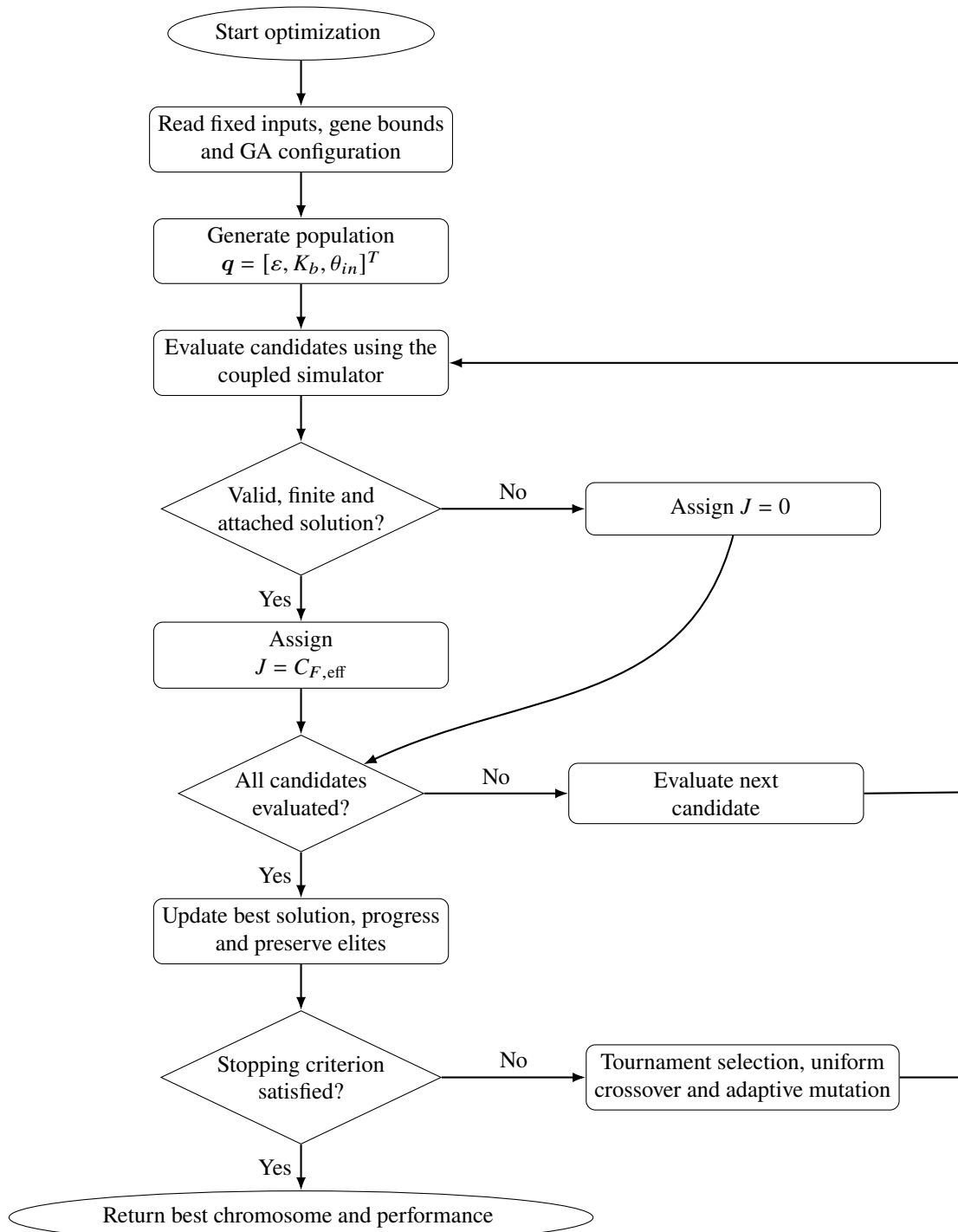


Figure 19: Flowchart of the implemented genetic-algorithm optimization.

10 Results and Discussion

10.1 Evaluation case and data provenance

The following results use two complete simulator exports retained with the software. Both cases use the same operating point and are evaluated with the BLIMP-lite closure, so the comparison changes only the three optimized design variables. The manually selected baseline is

$$\mathbf{q}_0 = [5.6, 0.80, 30^\circ]^T, \quad (245)$$

whereas the best geometry retained from an illustrative GA run is

$$\mathbf{q}^* = [5.071819, 0.600180, 39.30912^\circ]^T. \quad (246)$$

The latter run used a user-adjusted 40° upper bound for θ_{in} , rather than the current 35° default.

Table 3: Fixed operating-point and pre-sizing inputs used in the comparison.

Quantity	Symbol	Value
Chamber pressure	p_c	30 bar
Mixture ratio	O/F	6.5
Ambient pressure	p_a	0.841 atm
Throat radius	r_t	17.28 mm
Chamber diameter	D_c	120 mm
Convergent angle	θ_{sub}	50°
Reference cone half-angle	α	15°

The CSV files used by the plots below are included with the manuscript. Consequently, the figures are generated by PGFPlots from the numerical exports rather than inserted as screenshots. Since the saved run did not retain a random seed or complete population history, it is reported as a representative optimization result, not as evidence of statistical GA repeatability or guaranteed global optimality.

10.2 Geometry selected by the loss-corrected objective

Table 4: Baseline and retained GA geometry.

Quantity	Baseline	Retained GA candidate
Expansion ratio, ε	5.6000	5.0718
Bell fraction, K_b	0.8000	0.6002
Initial angle, θ_{in}	30.000°	39.309°
Derived exit angle, θ_{out}	3.790°	0.550°
Total axial length	121.886 mm	100.774 mm

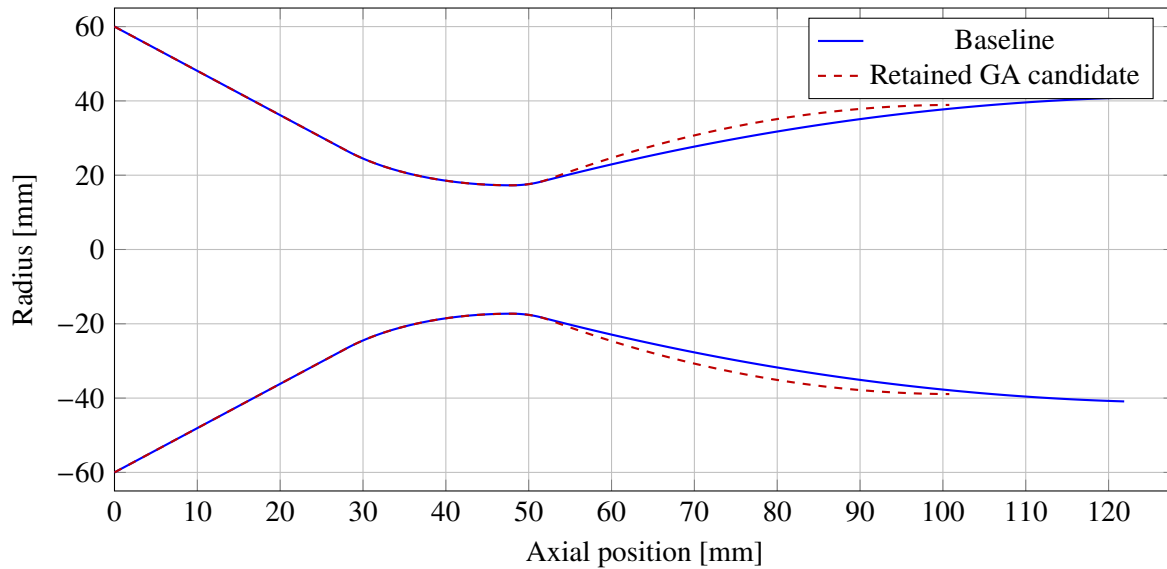


Figure 20: Baseline and retained GA contours generated directly from the exported coordinates.

The selected geometry is 17.32% shorter in total axial length. The higher initial angle creates stronger early expansion, while the quadratic closure reduces the derived exit angle to approximately 0.55° . This result must be read together with the model limitations: there is no independent MOC or shock-structure analysis penalizing rapid internal turning. The admissible bounds therefore remain an important part of the engineering definition of the search space.

10.3 Effective thrust coefficient and loss decomposition

Table 5: BLIMP-lite performance comparison at the specified ambient pressure.

Quantity	Baseline	Retained GA candidate
Exit pressure, p_e [bar]	0.7834	0.9024
Exit condition	Overexpanded	Underexpanded
$C_{F,mom}^{CEA}$	1.511664	1.490527
η_{div}	0.998906	0.999977
η_{BL}	0.996066	0.996610
$C_{F,p}$	-0.012839	0.008504
$C_{F,fric}$	0.017646	0.014309
$C_{F,eff}$	1.473586	1.479634
Effective thrust [N]	4147.0	4164.0

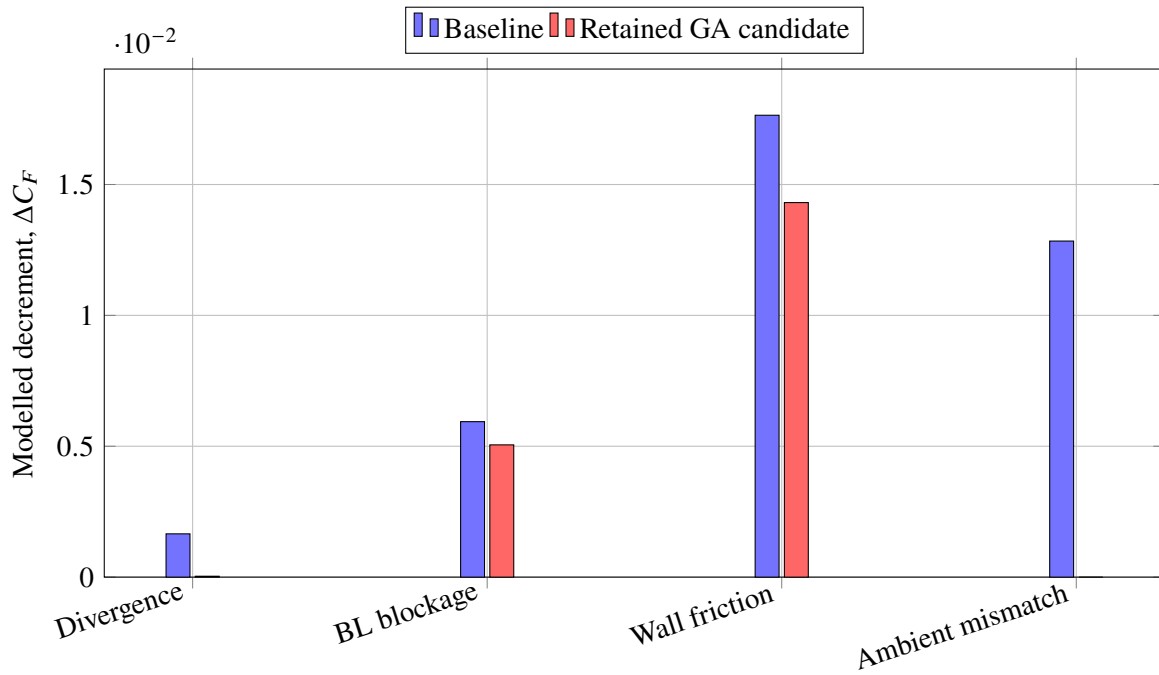


Figure 21: Non-negative loss contributions in thrust-coefficient units. A positive pressure-thrust contribution is a gain and is therefore excluded from the ambient mismatch bar.

The effective thrust coefficient increases from 1.473586 to 1.479634, a 0.41% improvement for fixed $p_c A_t$. The retained candidate does not simply choose the ideal pressure-matched expansion ratio $\varepsilon_{ideal} = 5.2792$. Instead, it accepts a small positive pressure-thrust contribution and reduces the modelled wall friction through the shorter contour. In the selected candidate, wall friction accounts for approximately 73.8% of the non-negative modelled loss, boundary-layer blockage for 26.1%, and exit divergence for less than 0.2%. These percentages describe the implemented closure and must not be interpreted as experimentally validated loss attribution.

10.4 Axial flow and boundary-layer response

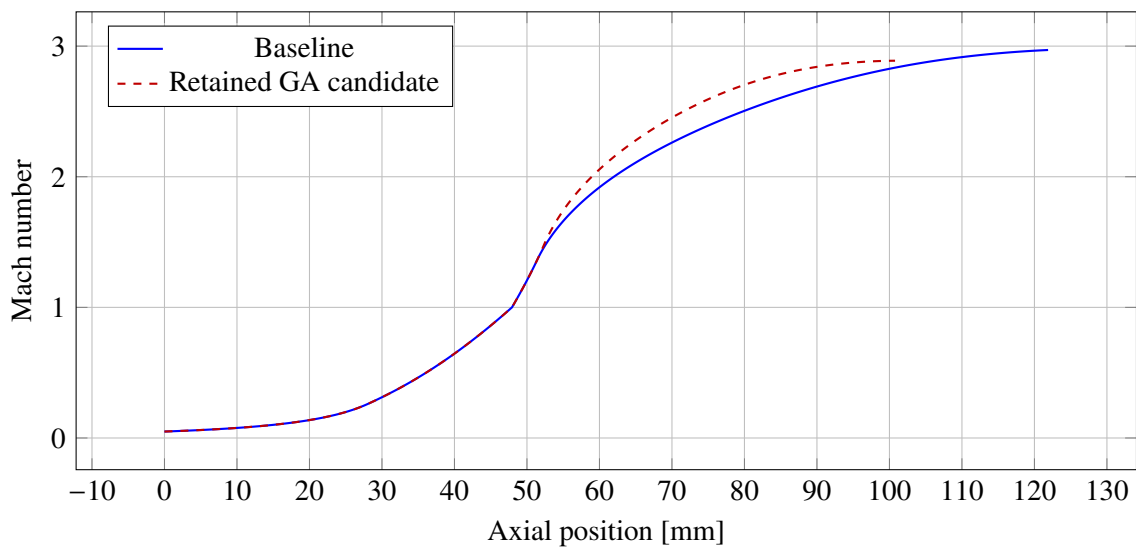


Figure 22: Quasi-one-dimensional Mach reconstruction for the two geometries.

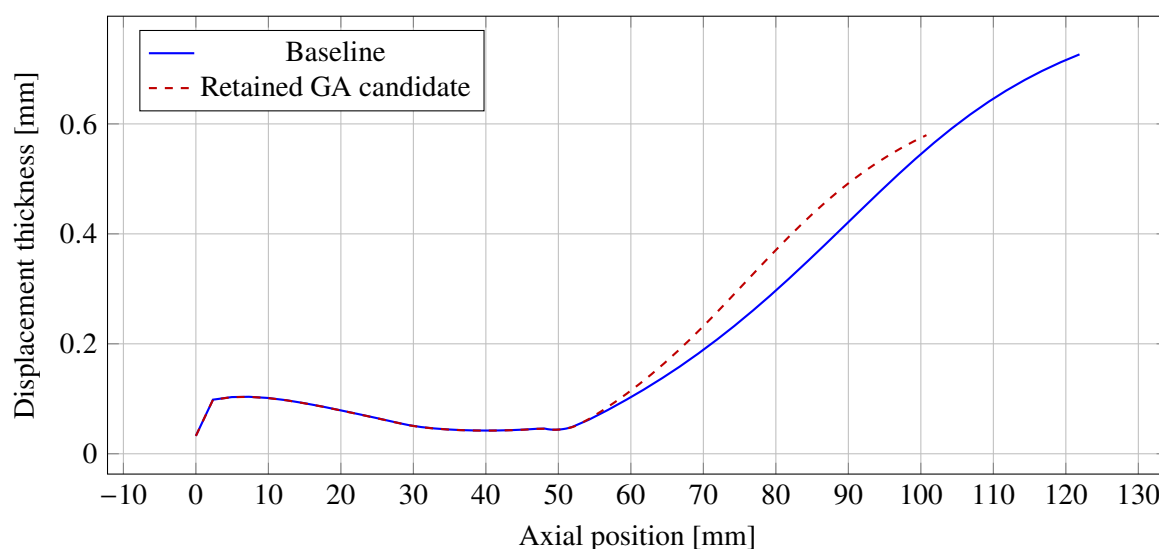


Figure 23: BLIMP-lite compressible displacement-thickness distributions.

The exported profiles remain finite and continuous across all four geometric segments. The comparison also illustrates why η_{BL} and $C_{F,fric}$ are kept separate: the former represents the effective-area velocity correction, whereas the latter is obtained by integrating the resolved wall shear. A publication-level assessment of optimizer robustness will additionally require repeated seeded runs, population convergence statistics and validation of the retained candidate with an independent higher-fidelity method.

11 Limitations and Future Work

The present framework is a reduced-order, single-operating-point preliminary design tool. Its principal limitations are:

- no method-of-characteristics or multidimensional wave analysis of the parameterized contour;
- no CFD-resolved shocks, separation or shock–boundary-layer interaction;
- no finite-rate chemistry or station-by-station equilibrium/frozen transition;
- interpolated RocketCEA properties used with local perfect-gas relations;
- fully turbulent, attached-flow BLIMP-lite and Quick closures with an engineering inlet boundary-layer initialization;
- an adiabatic wall with no conjugate thermal, coolant, ablative or structural model;
- no experimental calibration of η_{div} , η_{BL} or $C_{F,fric}$;
- a single objective and a single ambient pressure per optimization run.

Future work should prioritize:

1. repeated seeded GA runs with convergence and uncertainty statistics;
2. comparison of Quick and BLIMP-lite rankings over a common design sample;
3. independent validation of geometry continuity, boundary-layer quantities and thrust loss against published experiments or mesh-refined RANS/CFD;
4. trajectory-weighted or multi-point optimization across ambient pressure;
5. thermal and structural constraints for manufacturable designs;
6. optional MOC analysis of selected parametric contours as a validation layer, without replacing the present geometry family;
7. comparison between conical, Rao TOP and optimized parametric bell contours.

The lack of a separate empirical turning penalty is deliberate. Rapid turning is currently controlled through geometric admissibility and bounded values of θ_{in} , while its multidimensional wave consequences remain outside the validity of the quasi-one-dimensional closure. Such effects should be introduced only through a defensible MOC, CFD or experimental analysis rather than an uncalibrated coefficient.

12 Conclusion

This work presents an implemented preliminary-design framework for a single bell nozzle. A continuous piecewise contour composed of a straight convergent, circular subsonic and supersonic throat arcs, and a quadratic divergent section supports two-dimensional plotting, axisymmetric three-dimensional visualization, CAD export and quasi-one-dimensional flow reconstruction. The throat radius is retained from engine pre-sizing, while the exit radius follows from the expansion ratio.

The genetic algorithm varies three real-valued design variables: ε , K_b and θ_{in} . The convergent angle and reference cone half-angle remain user-selected fixed inputs, and the exit wall angle is derived from the resulting quadratic contour. The optimized objective is the effective ambient thrust coefficient, combining RocketCEA momentum performance, exit divergence, compressible boundary-layer blockage, signed pressure thrust and integrated wall friction. No independent curvature or empirical turning efficiency is used.

Two viscous evaluation levels provide a practical cost–fidelity trade-off. Quick screening uses a weak momentum-integral closure, whereas BLIMP-lite resolves a wall-normal velocity profile with a Cebeci–Smith eddy-viscosity model and is used for final reporting. Both assume an attached, fully turbulent and adiabatic wall; hence $T_w = T_{aw}$ and the present model does not predict non-zero wall heat flux.

For the representative saved run, the selected geometry reduced total length by 17.32% and increased $C_{F,eff}$ by 0.41% relative to the baseline. This result demonstrates the coupled workflow rather than final nozzle qualification. The framework is therefore suitable for transparent preliminary comparison and design-space exploration, provided that final candidates are validated using independent higher-fidelity analysis and experimental evidence.

References

- [1] G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 9th ed., Wiley, 2017.
- [2] S. Gordon and B. J. McBride, *Computer Program for Calculation of Complex Chemical Equilibrium Compositions and Applications*, NASA Reference Publication 1311, 1994.
- [3] C. E. DeLong, *RocketCEA Documentation*, Python interface for NASA CEA.
- [4] J. D. Anderson, *Modern Compressible Flow: With Historical Perspective*, McGraw-Hill.
- [5] R. M. Evans, *BLIMP-J User's Manual: Boundary Layer Integral Matrix Procedure for JANNAF Rocket Engine Performance Evaluation Methodology*, AEROTHERM-UM-75-64, NASA-CR-144046, 1975.
- [6] W. S. Bonnett and R. M. Evans, *Boundary Layer Integral Matrix Procedure: Verification of Models*, AEROTHERM-77-239, NASA-CR-150271, 1977.
- [7] T. Cebeci and A. M. O. Smith, *Analysis of Turbulent Boundary Layers*, Academic Press, 1974.
- [8] A. Walz, *Boundary Layers of Flow and Temperature*, MIT Press, 1969.
- [9] D. R. Bartz, "A Simple Equation for Rapid Estimation of Rocket Nozzle Convective Heat Transfer Coefficients," *Jet Propulsion*, vol. 27, pp. 49–51, 1957.
- [10] A. F. Gad, "PyGAD: An Intuitive Genetic Algorithm Python Library," *Multimedia Tools and Applications*, 2023.